

Intelligent Decision Support System for Sri Lankan Tea Estate Management

R26-IT-118

Project Proposal Report

Gunasekara G.N.D (IT22279620)

B.Sc. (Hons) Degree in Information Technology

Department of information technology

Sri Lanka Institute of Information Technology

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Supervised By: Mrs. Uthpala Samarakoon

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DECLARATION

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I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Gunasekara G.N.D	IT22279620	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Uthpala Samarakoon	19/03/2026	

ABSTRACT

Weather conditions play a major role in tea plantation productivity, quality, and disease occurrence in Sri Lanka. Traditional weather monitoring depends on generic weather applications that provide basic forecasts without tea-specific guidance. Estate managers must manually check weather data and decide on cultivation activities like harvesting, irrigation, fertilization, and disease prevention. This manual approach takes too much time, responds to problems after they happen instead of preventing them, and often leads to delayed responses to critical weather events.

This research proposes a Tea-Specific Weather and Climate Intelligence System as a core component of an Integrated AI-Driven Intelligent Decision Support System for tea estate management. The system uses machine learning methods including LSTM (Long Short-Term Memory) networks and Random Forest models to analyze historical weather patterns from 10 years of data and real-time weather forecasts from APIs like Google Weather and OpenWeather.

The system automatically translates generic weather data into tea cultivation impacts. It predicts optimal harvest windows, disease outbreak risks, irrigation requirements, and quality forecasts. The system provides automated daily briefings at 6:00 AM with prioritized cultivation tasks, critical weather alerts 2-7 days in advance with detailed action plans, and annual strategic cultivation calendars based on historical climate patterns.

The system goes beyond basic weather forecasting by including predictive analytics for blister blight and red spider mite outbreaks based on temperature-humidity patterns, precision irrigation scheduling using evapotranspiration calculations, and harvest quality window identification that predicts tea leaf biochemical composition based on weather conditions.

By combining real-time weather intelligence, historical pattern analysis, and tea-specific cultivation recommendations, this research contributes a complete and proactive weather management solution designed specifically for Sri Lankan tea estates. The proposed approach helps with early intervention, reduces weather-related crop losses, optimizes resource usage, and supports sustainable, data-driven estate management practices.

Keywords: Weather Intelligence, Tea Cultivation, LSTM, Machine Learning, Irrigation Scheduling, Disease Risk Prediction, Decision Support System, Sri Lankan Tea Estates

Table of Contents

DECLARATION	1
ABSTRACT.....	2
LIST OF FIGURES	4
LIST OF TABLES	4
LIST OF APPENDICES	4
1. INTRODUCTION.....	5
1.1 Background and Context.....	5
1.2 Literature Review	7
1.3 Research Gap	9
2. OBJECTIVES	11
2.1 Main Objective	11
2.2 Specific Objectives	11
3. METHODOLOGY	12
3.1 Research Approach.....	12
3.2 Technical Methods and Algorithms	12
3.3 Tools and Platforms.....	14
3.4 Data Collection	15
3.5 Validation Strategy	16
4. HIGH-LEVEL SYSTEM ARCHITECTURE.....	18
4.1 Overview of the Proposed System	18
4.2 System Components.....	18
4.3 Component Interaction.....	21
4.4 Data Flow Explanation	23
5. USER REQUIREMENTS	24
5.1 Stakeholder Identification	24
5.2 Functional Requirements	25
5.3 Non-Functional Requirements.....	26
6. COMMERCIALIZATION PLAN	27
6.1 Target Market	27
6.2 Value Proposition.....	27
6.3 Pricing Strategy.....	28
6.4 Competitive Advantage	30
7. BUDGET AND JUSTIFICATION.....	31
8. WORK BREAKDOWN STRUCTURE.....	33

9. GANTT CHART	36
10. REFERENCES.....	37
11. APPENDICES	38
11.1 Survey Instrument	38
11.2 Detailed Diagrams.....	41
11.3 Preliminary UI Mockups.....	44
11.4 Risk Analysis	47
11.5 Ethical Documents	49

LIST OF FIGURES

Figure 4.1: Tea Weather Intelligence System Architecture	20
Figure 4.2: Component Interaction Workflow Diagram.....	21
Figure 4.3: User Interaction Flow Diagram	22
Figure 4.4: Continuous Learning and Model Improvement Cycle	23
Figure 9.1: Project Gantt Chart (March – November 2026).....	36

LIST OF TABLES

Table 1.1: Comparison of Existing Research Methods and Proposed Method.....	10
Table 6.1: Pricing Plans and Features	28
Table 7.1Budget table	31
Table 11.4.1: Project Risk Analysis.....	47

LIST OF APPENDICES

Figure 11.2.1: LSTM Neural Network Workflow for Weather Prediction.....	42
Figure 11.2.2: Random Forest Disease Prediction Model	42
Figure 11.2.3: System Data Flow Diagram.....	43
Figure 11.2.4: Automated Daily Briefing Generation Workflow	44
Figure 11.3.1: Web Dashboard Home Page.....	45
Figure 11.3.2: Mobile Daily Briefing Interface	46

1. INTRODUCTION

The tea industry is one of the most important contributors to Sri Lanka's national economy. It provides employment opportunities, foreign exchange earnings, and supports rural livelihoods. Maintaining high tea quality and consistent production is critical for sustaining global competitiveness. However, tea leaf diseases and weather variability remain major challenges affecting yield, leaf quality, and overall plantation productivity. Common diseases such as blister blight, red rust, and grey blight can spread rapidly under favorable environmental conditions, causing substantial crop losses if not detected and treated at an early stage.

Weather directly impacts every aspect of tea cultivation including growth rates, plucking schedules, disease occurrence, pest outbreaks, irrigation needs, and final tea quality. Unpredictable rainfall, temperature extremes, humidity fluctuations, and seasonal variations can cause substantial crop losses and quality degradation if not anticipated and managed effectively.

In most Sri Lankan tea estates, weather monitoring primarily depends on checking generic weather applications or websites designed for general public use. Estate managers manually check this basic weather information and make cultivation decisions based on experience and intuition. This traditional approach has several critical limitations. It is time-consuming, taking 2-3 hours daily. It is reactive rather than proactive. It lacks tea-specific interpretation. It provides no disease risk predictions. It offers no quality forecasting. Many critical weather alerts are missed due to manual checking.

Advancements in Artificial Intelligence and Machine Learning, particularly time-series forecasting algorithms like LSTM networks and ensemble methods like Random Forest, have demonstrated strong capabilities in weather pattern recognition and agricultural prediction tasks. However, most existing weather systems provide only generic forecasts without crop-specific intelligence or actionable cultivation recommendations.

This research proposes a Tea-Specific Weather and Climate Intelligence System as a component of an Integrated AI-Driven Intelligent Decision Support System for Sri Lankan tea estates. The system collects weather data from multiple APIs including Google Weather, OpenWeather, and Visual Crossing. It analyzes 10 years of historical climate patterns and uses machine learning models to predict tea-specific cultivation impacts. The system automatically generates daily cultivation briefings at 6:00 AM with prioritized tasks. It provides 2-7 day advance warnings for critical weather events with detailed action plans. It predicts disease outbreaks based on weather patterns for blister blight and red spider mite. It calculates precise irrigation requirements using evapotranspiration models. It identifies optimal harvest quality windows and creates annual strategic cultivation calendars.

By combining real-time weather intelligence with predictive pattern analysis and automated decision support features, the proposed solution aims to enhance early diagnosis of weather risks, reduce crop losses, optimize water and chemical usage, improve tea quality through better timing, and increase overall estate management efficiency. This research contributes toward a sustainable and data-driven approach to tea weather management tailored specifically to the Sri Lankan tea industry.

1.1 Background and Context

Tea is one of Sri Lanka's most valuable agricultural exports. It contributes significantly to national income, rural employment, and foreign exchange earnings. Sri Lanka produces approximately 300

million kilograms of tea annually, making it one of the world's leading tea exporters. Maintaining high leaf quality and stable yield is essential for sustaining the country's global reputation in premium tea production.

However, weather variability and climate patterns pose major threats to plantation productivity and tea quality. Temperature fluctuations affect growth rates and biochemical composition of tea leaves. Rainfall patterns determine soil moisture, harvest scheduling, and disease occurrence. Humidity levels trigger fungal diseases like blister blight. Solar radiation impacts photosynthesis and quality compound formation. Wind conditions affect evapotranspiration rates and irrigation needs.

Common weather-related challenges in Sri Lankan tea estates include unpredictable monsoon patterns. The Southwest monsoon from May to September and Northeast monsoon from December to February bring heavy rainfall that can delay harvesting, cause waterlogging, trigger disease outbreaks, and reduce tea quality. Temperature extremes also cause problems. Heat waves above 30°C cause heat stress, reduce growth rates, increase pest activity like red spider mites, and decrease tea quality. Cold spells below 15°C slow growth and can damage young shoots. Drought periods with extended dry spells lead to moisture stress, reduced yield, increased irrigation costs, and poor flush quality with bitter taste. Humidity-related diseases occur when high humidity above 85% for extended periods triggers blister blight outbreaks that can devastate entire plantations if not detected early. Sudden weather changes like unexpected thunderstorms, hail, or frost events can cause immediate crop damage and disrupt planned cultivation activities.

Currently, weather monitoring in many tea estates relies primarily on checking generic weather applications like OpenWeather, AccuWeather, or Weather.com, or listening to local TV and radio forecasts. This traditional method has several critical limitations. It is time-consuming, with estate managers spending 2-3 hours daily checking multiple weather sources, comparing forecasts, and trying to interpret what the weather means for tea cultivation. It is reactive not proactive, with managers responding to weather after it happens rather than preparing in advance. For example, discovering disease-favorable conditions only after outbreak begins. It has no tea-specific intelligence. Generic weather apps say "25°C, 80% humidity, 10mm rain" but don't explain if this is good or bad for tea, what actions to take, or how it affects quality. It has no historical context. Current weather forecasts don't compare to previous years or identify similar patterns that led to good or bad outcomes in the past. Missed alerts occur because manual checking means many critical weather warnings are missed, especially overnight or during busy periods. There is no integrated planning. Weather information is separate from cultivation planning, so managers must manually connect weather forecasts to irrigation schedules, harvest timing, disease prevention, and other activities.

As a result, estate managers, field supervisors, plantation workers, and tea factory operators are directly affected by poor weather intelligence and delayed management responses. Late responses to bad weather lead to crop losses averaging 15-25% during critical events.

This research aligns with the Sustainable Development Goals. It supports SDG 2 for Zero Hunger through improved agricultural productivity. It supports SDG 9 for Industry, Innovation and Infrastructure. It supports SDG 13 for Climate Action through climate-smart agriculture. The study belongs to the Technology Integration and Management research cluster, focusing on integrating Artificial Intelligence technologies into practical plantation management systems.

1.2 Literature Review

Artificial Intelligence and machine learning technologies have increasingly been applied in agriculture to improve weather forecasting, crop monitoring, and climate-smart decision-making. Time-series analysis using deep learning models has become one of the most effective techniques for predicting agricultural outcomes based on weather patterns.

Haciimamoglu et al. (2025) assessed climate change effects on Turkish tea farming using machine learning models [1]. Their research showed that temperature and precipitation significantly affect tea yield. Machine learning models can predict productivity based on climate variables. However, the study focused on long-term climate trends rather than short-term operational weather intelligence for daily cultivation decisions.

Ali et al. (2022) proposed a hybrid approach combining simulation models with machine learning for tea crop yield prediction [2]. They showed that weather-based simulation models achieve 75-82% accuracy in predicting tea yield. However, their system required extensive field data collection and did not provide real-time cultivation recommendations or disease risk predictions.

Xing et al. (2024) predicted changes in suitable tea habitats in China using machine learning models based on climate, terrain, and soil variables [3]. Their research identified optimal climate ranges for tea cultivation with temperature at 15-25°C and precipitation at 1000-2000mm annually. While this work demonstrated the importance of climate factors, it focused on habitat suitability rather than operational weather intelligence.

Zhang et al. (2023) evaluated suitable areas for premium teas in Yunnan using machine learning models [4]. They found that temperature and precipitation are critical factors affecting tea quality, with optimal conditions being 17-22°C and 1,100mm plus annual precipitation. However, their analysis was at a regional scale and did not address day-to-day weather decision support.

Lou et al. (2021) analyzed the impact of climate change on tea output in Zhejiang, China [6]. They found that temperature and precipitation variations significantly affect inter-annual tea yield. Their statistical analysis showed strong correlations between weather patterns and productivity but lacked predictive forecasting capabilities.

Zhang et al. (2023) developed ensemble machine learning algorithms for estimating daily evapotranspiration in tea plantations [7]. They tested six meteorological data scenarios and found that Random Forest and bagging ensemble models achieved high accuracy with R-squared greater than 0.85 in predicting evapotranspiration rates. This research demonstrated the feasibility of machine learning-based irrigation scheduling but did not integrate broader cultivation intelligence.

Mallik and Ghosh (2022) studied the impact of climate on tea production in the Dooars region of India [8]. They identified that monsoon timing, temperature variations, and rainfall distribution critically affect tea yield and quality. However, their analysis was looking backwards and did not develop predictive decision support tools.

Wijeratne (1996) analyzed the vulnerability of Sri Lankan tea production to global climate change [9]. The research noted that increased temperature and changed rainfall patterns increase disease risks, particularly blister blight and red spider mites. The research emphasized the need for climate-adaptive management but did not propose AI-driven solutions.

Biggs et al. (2018) studied the tea landscape of Assam under changing climate [10]. They highlighted that unpredictable weather patterns increase pest and disease pressure. Their multi-stakeholder research showed that farmers need better weather-based early warning systems for sustainable livelihoods.

Cheserek et al. (2015) analyzed links between climate variables and tea production in Kenya [11]. They found that temperature, rainfall, and sunshine hours are strongly related to tea yield and quality. They recommended developing weather-based advisory systems but did not implement such systems.

Several commercial weather APIs provide agricultural weather data. OpenWeather Agro API offers soil temperature, moisture indices, accumulated temperature, and cumulative precipitation tracking [12]. However, it provides raw weather data without crop-specific interpretation or cultivation recommendations. Visual Crossing Weather API provides historical weather data spanning 40 plus years with climate analytics [13]. While useful for pattern analysis, it lacks real-time tea-specific intelligence and automated decision support. Tomorrow.io Weather API offers 60 plus data layers including pollen, air quality, and precipitation forecasts [14]. However, these are generic parameters not translated into tea cultivation impacts. The Weather Company from IBM provides enterprise-grade weather APIs with real-time and historical data insights [15]. Their platform focuses on general agricultural weather without tea-specific disease models or quality forecasting.

While existing studies demonstrate the strong impact of weather on tea cultivation and the potential of machine learning for agricultural forecasting, several critical gaps exist. There is a lack of tea-specific weather intelligence. Most weather systems provide generic forecasts like temperature, rainfall, and humidity without translating these into tea cultivation impacts like harvest timing, disease risk, and quality effects. There are no proactive cultivation recommendations. Existing research focuses on looking at past relationships and yield prediction but does not generate automated daily operational guidance on what tasks to do today based on weather. There is limited disease risk forecasting. Although studies know about weather-disease relationships, few systems predict outbreaks 3-7 days in advance with preventive action plans for tea-specific diseases like blister blight and red spider mite. There is no integrated decision support. Weather data, disease prediction, irrigation scheduling, and quality forecasting are studied separately but not brought together into unified decision support platforms for estate managers. There is an absence of historical pattern learning. Most systems use current or short-term forecasts only, missing the opportunity to learn from 10 plus years of weather-outcome patterns to improve predictions. There is a lack of real-world implementation. Academic research shows machine learning model accuracy in controlled experiments but rarely implements end-to-end systems deployed in actual tea estates with user interfaces and operational workflows.

Therefore, there is a clear need for a complete AI-driven solution that goes beyond simple weather forecasting to provide tea-specific cultivation intelligence, proactive disease warnings, quality optimization, and integrated estate management support.

1.3 Research Gap

Although recent studies have shown the effectiveness of machine learning techniques such as LSTM networks and Random Forest algorithms for agricultural weather forecasting, most existing approaches focus primarily on yield prediction or climate suitability analysis. For example, several studies successfully predict tea productivity based on temperature and rainfall data with reasonable accuracy, but their systems are typically limited to looking at relationships only.

Furthermore, existing research rarely adds decision-making capabilities such as disease outbreak prediction, harvest quality window identification, precision irrigation scheduling, or automated cultivation task generation. As a result, current systems provide limited support for proactive weather-based management in real plantation environments. Estate managers still depend heavily on manual interpretation of generic weather forecasts to determine daily cultivation strategies.

Table 1.1 below compares existing research methods and systems with the proposed method. The table shows what technologies and methods each approach has implemented.

According to Table 1.1, existing research and commercial systems provide weather data collection and some basic features, but none offer the complete set of tea-specific intelligence features proposed in this system. The main gaps include no tea-specific weather translation where existing systems say "25°C, 80% humidity" but don't explain if this is optimal for plucking, risky for disease, or good for quality. Systems are reactive rather than proactive, with current approaches responding to weather after it impacts cultivation, rather than predicting issues 3-7 days in advance with prevention strategies. There is missing disease-weather integration, with weather and disease management being separate areas and no system predicting blister blight or mite outbreaks based on temperature-humidity patterns. There is no quality forecasting, with existing research not predicting how current weather will affect biochemical composition and market grade of harvested tea. There is lack of automated operational guidance, with estate managers still manually interpreting forecasts and planning daily tasks and no system providing prioritized action lists. There is limited long-term intelligence, with most systems focusing on 7-16 day forecasts without using 10 plus years of historical weather-outcome data for pattern recognition.

Therefore, there is a need for a more complete AI-driven solution that goes beyond simple weather forecasting. The proposed research addresses this gap by developing a tea-specific LSTM-based weather intelligence system that brings together weather data collection, historical pattern analysis, disease risk prediction, quality forecasting, irrigation optimization, and automated cultivation recommendations to support intelligent and proactive tea plantation management.

Table 1.1: Comparison of Existing Research Methods and Proposed Method

Technologies and Methods	Haciimamoglu [1]	Ali et al. [2]	Zhang et al. [7]	OpenWeather Agro [12]	Proposed System
Weather data collection	Yes	Yes	Yes	Yes	Yes
Historical pattern analysis (10+ years)	Yes	No	No	Yes	Yes
Tea-specific impact interpretation	No	No	No	No	Yes
Disease outbreak prediction (3-7 days advance)	No	No	No	No	Yes
Harvest quality window identification	No	No	No	No	Yes
Precision irrigation scheduling	No	No	Basic	Basic	Advanced
Automated daily cultivation briefings	No	No	No	No	Yes
Critical weather alerts with action plans	No	No	No	Alerts only	Alerts plus Actions
Annual strategic cultivation calendar	No	No	No	No	Yes
Real-time web or mobile system	No	No	No	API only	Full application
Integration with estate management	No	No	No	No	Yes

2. OBJECTIVES

2.1 Main Objective

To design and evaluate an intelligent machine learning-based Tea-Specific Weather and Climate Intelligence System that collects and analyzes weather data from multiple APIs, predicts tea cultivation impacts using LSTM and Random Forest models, and provides proactive decision support through automated daily briefings, disease outbreak alerts, harvest quality forecasts, and precision irrigation scheduling to support data-driven and sustainable weather management in Sri Lankan tea plantations.

2.2 Specific Objectives

The research has seven specific objectives. First, to collect and combine weather data from multiple sources including Google Weather API, OpenWeather API, and Visual Crossing Weather API, along with 10 years of historical climate records from 2015 to 2025 for Nuwara Eliya, Kandy, and Ratnapura tea regions. This creates a complete weather database for tea-specific analysis. Second, to design and implement an LSTM neural network model that can analyze time-series weather patterns and predict tea cultivation impacts including growth rates, disease risks, quality scores, and irrigation requirements based on temperature, humidity, rainfall, and solar radiation data.

Third, to develop a disease outbreak prediction module using pattern recognition algorithms that identify weather conditions favorable for blister blight when there are cool temperatures plus high humidity and red spider mite when there are hot temperatures plus low humidity. The system will provide 3-7 day advance warnings with confidence scores and preventive action plans. Fourth, to implement a harvest quality forecasting system that analyzes weather patterns including temperature range, sunshine hours, rainfall distribution, and day-night temperature differences to predict biochemical composition of tea leaves including polyphenols, amino acids, and caffeine. This helps identify optimal harvest windows for premium grade production.

Fifth, to create a precision irrigation scheduling system using evapotranspiration models based on the Penman-Monteith equation combined with weather forecasts. This calculates exact irrigation timing and water volume requirements for different tea varieties including TRI-2025 and TRI-3055, reducing water waste by 25-35%. Sixth, to combine the weather intelligence system into a web-based dashboard application with automated daily briefings delivered at 6:00 AM, critical weather alerts with 2-hour advance warnings, weekly planning summaries, and annual cultivation calendars. This enables estate managers and supervisors to access proactive cultivation recommendations in real-time.

Seventh, to analyze historical weather-outcome data from 10 years to identify recurring climate patterns, seasonal trends, and unusual events that affected tea yield and quality. This supports predictive decision-making and long-term strategic planning for climate-smart tea estate management.

3. METHODOLOGY

3.1 Research Approach

This research follows the Design Science Research methodology, which focuses on designing, developing, and evaluating an innovative technological solution to address a real-world problem. In this study, the artifact developed is a Tea-Specific Weather and Climate Intelligence System that supports proactive weather-based decision-making for tea plantation management. The research process includes six main phases.

Phase 1 involves problem identification and requirements analysis through conducting stakeholder interviews with estate managers, supervisors, and tea experts, identifying current weather monitoring practices and pain points, and defining functional and non-functional requirements for the system. Phase 2 covers literature review and technology selection by reviewing existing weather forecasting systems and agricultural AI applications, identifying research gaps in tea-specific weather intelligence, selecting appropriate machine learning algorithms including LSTM and Random Forest, and evaluating weather API options including Google Weather, OpenWeather, and Visual Crossing.

Phase 3 handles data collection and preparation through collecting 10 years of historical weather data from 2015 to 2025 for tea regions, gathering tea cultivation outcome data including yield records, quality grades, and disease incidents, cleaning and processing data by handling missing values and normalizing ranges, and creating labeled training datasets linking weather patterns to outcomes. Phase 4 focuses on model development and training by designing LSTM architecture for time-series weather pattern recognition, implementing Random Forest models for disease risk classification, training models using 70% training, 15% validation, and 15% testing split, and optimizing hyperparameters including learning rate, number of layers, and epochs.

Phase 5 covers system integration and implementation through developing weather data collection pipeline with API integration, building real-time prediction engine using trained machine learning models, creating automated briefing generation system, designing web-based dashboard interface using React frontend and Node.js backend, and implementing alert notification system with email, SMS, and push notifications. Phase 6 involves evaluation and validation by testing system accuracy using validation datasets, deploying pilot system in 2-3 tea estates for real-world testing, collecting user feedback from estate managers and supervisors, measuring performance metrics including prediction accuracy and response time, and comparing system recommendations versus manual expert decisions.

The proposed system uses deep learning and ensemble machine learning techniques to analyze weather data and automatically predict tea-specific cultivation impacts. The system is designed as a cloud-based web application that continuously monitors weather APIs, analyzes patterns using trained machine learning models, and generates automated recommendations delivered through email, mobile notifications, and web dashboards.

3.2 Technical Methods and Algorithms

The core technical methods used in this research combine time-series forecasting, classification algorithms, and optimization techniques. LSTM or Long Short-Term Memory Networks are used for

weather pattern recognition. LSTM is a type of recurrent neural network specifically designed for sequence data and time-series analysis. Unlike traditional neural networks, LSTM can remember long-term patterns and dependencies in sequential data, making it ideal for weather forecasting.

LSTM is used for weather intelligence because it remembers seasonal patterns like monsoon cycles and dry periods. It captures long-term climate trends from 10-year historical patterns. It predicts future states based on past sequences and handles irregular time intervals and missing data points. The LSTM architecture for this system has an input layer for weather sequences using 7-day windows. LSTM Layer 1 has 128 neurons and learns short-term patterns. LSTM Layer 2 has 64 neurons and learns medium-term patterns. LSTM Layer 3 has 32 neurons and learns long-term patterns. A dense layer with 16 neurons combines patterns. The output layer provides multiple predictions including growth rate impact showing positive or negative percentage, disease risk score from 0 to 100%, quality score from 1 to 5 stars, and irrigation need in millimeters of water.

The training process takes input of 7 days of historical weather including temperature, humidity, rain, wind, and sun. It produces output of next 7 days cultivation impact predictions. It uses Mean Squared Error as the loss function for regression tasks. It uses Adam optimizer with learning rate 0.001. It runs for 50-100 epochs with early stopping to prevent overfitting.

Random Forest is used for disease outbreak classification. Random Forest is an ensemble learning method that combines multiple decision trees to make predictions. It is highly effective for classification tasks and handles complex non-linear relationships between weather variables and disease occurrence. Random Forest is used for disease prediction because it handles multiple input variables including temperature, humidity, rain, and wind. It identifies which weather factors most influence disease risk. It is strong against outliers and noisy data and provides confidence scores for predictions.

The system has disease classification models for blister blight prediction and red spider mite prediction. For blister blight, input features include average temperature from past 3 days, minimum temperature from morning readings, average humidity from past 3 days, maximum humidity from morning peaks, rainfall amount from past 7 days, and wind speed which affects spore dispersal. The decision tree logic works as follows. If temperature is 15-20°C and humidity is greater than 90% for 3 days and there is light rain, then there is high risk with 85% confidence. Else if temperature is 15-20°C and humidity is greater than 85% for 2 days, then there is medium risk with 60% confidence. Otherwise there is low risk with 25% confidence. The output shows risk level as low, medium, or high plus confidence score from 0 to 100%.

For red spider mite prediction, input features include maximum temperature from past 5 days, average humidity from past 5 days, days without rainfall, and evapotranspiration rate. The decision tree logic works as follows. If temperature is greater than 30°C for 5 days and humidity is less than 60% and no rain for 7 plus days, then there is high risk with 90% confidence. Else if temperature is greater than 28°C for 3 days and humidity is less than 65%, then there is medium risk with 55% confidence. Otherwise there is low risk with 20% confidence.

Random Forest training uses 500 trees in the ensemble of 500 decision trees. Maximum depth is 10 to prevent overly complex trees. Features per split uses square root of total features. Training data comes from 10 years of weather plus disease occurrence records. Class balancing handles imbalanced data with more "no disease" days than "disease" days.

Evapotranspiration calculation for irrigation scheduling uses the FAO Penman-Monteith equation, which is the standard method for calculating crop water requirements. The reference evapotranspiration ET_0 equals $0.408 \times \Delta \times \frac{R_n - G + \gamma \times 900}{T + 273} \times \frac{e_s - e_a}{\Delta + \gamma \times (1 + 0.34 \times u_2)}$. In this equation, ET_0 is reference evapotranspiration in millimeters per day. R_n is net radiation in megajoules per meter squared per day from solar radiation data. G is soil heat flux, usually 0 for daily calculations. T is mean temperature in degrees Celsius. u_2 is wind speed at 2 meter height in meters per second. e_s is saturation vapor pressure in kilopascals. e_a is actual vapor pressure in kilopascals from humidity. Δ is slope of vapor pressure curve. γ is psychrometric constant.

Crop-specific adjustment calculates crop evapotranspiration ET_c as ET_0 times K_c , where ET_c is crop evapotranspiration for tea and K_c is crop coefficient with value 1.0 for tea, similar to grass reference. Irrigation need calculation determines irrigation need as ET_c minus effective rainfall plus soil moisture deficit. For example, if ET_0 equals 4.5 millimeters per day calculated from weather data, then ET_c equals 4.5×1.0 equals 4.5 millimeters per day. If effective rainfall equals 2.0 millimeters from yesterday's rain at 70% effectiveness and soil moisture deficit equals 10 millimeters with current at 58% and optimal at 68%, then irrigation need equals $4.5 - 2.0 + 10$ equals 12.5 millimeters to apply today.

The quality scoring algorithm combines multiple weather factors with weighted importance based on tea science research. Quality score from 1 to 5 stars equals $0.30 \times \text{temperature score} + 0.25 \times \text{sunshine score} + 0.20 \times \text{rainfall score} + 0.15 \times \text{humidity score} + 0.10 \times \text{day-night differential score}$. Temperature score gives 5 stars for 20-25°C as optimal, 4 stars for 18-20°C or 25-27°C as good, 3 stars for 15-18°C or 27-30°C as acceptable, and 2 stars for less than 15°C or greater than 30°C as poor. Sunshine score gives 5 stars for 6-8 hours per day as optimal, 4 stars for 5-6 or 8-9 hours as good, 3 stars for 4-5 or 9-10 hours as acceptable, and 2 stars for less than 4 or greater than 10 hours as poor. Rainfall score gives 5 stars for 0-5 millimeters per day with no rain during harvest, 4 stars for 5-10 millimeters per day with morning rain only, 3 stars for 10-20 millimeters per day, and 2 stars for greater than 20 millimeters per day requiring skip harvest. Final quality interpretation shows 4.5-5.0 stars as excellent for FBOP grade, 3.5-4.4 stars as good for BOP grade, 2.5-3.4 stars as acceptable for Pekoe grade, 1.5-2.4 stars as poor for low grade, and less than 1.5 stars as very poor requiring skip harvest.

3.3 Tools and Platforms

Programming languages used include Python version 3.9 or higher for core machine learning development and data processing, JavaScript with Node.js for backend API services, and React.js for frontend user interface. Machine learning frameworks include TensorFlow version 2.x for LSTM neural network implementation, Keras as high-level neural network API, Scikit-learn for Random Forest and data preprocessing, and NumPy and Pandas for data manipulation and analysis.

Weather data sources include Google Weather API for real-time current conditions, OpenWeather API for 16-day forecasts plus historical data, Visual Crossing Weather API for 40 plus years historical climate data, and NOAA Climate Data Online for long-term climate records. Backend technologies include Node.js with Express for RESTful API server, PostgreSQL as relational database for structured data, MongoDB as document database for JSON weather data, and Redis as caching layer for fast API responses.

Frontend technologies include React.js as component-based UI framework, Material-UI as modern UI component library, Chart.js or Recharts for data visualization including graphs and charts, and Axios as HTTP client for API requests. Cloud and deployment uses AWS EC2 or Google Cloud for cloud hosting of application, AWS S3 for storage of historical weather data, Docker for containerization for deployment, and Nginx as web server and reverse proxy.

Notification services include Twilio for SMS alert delivery, SendGrid for email notification system, and Firebase Cloud Messaging for mobile push notifications. Development tools include Jupyter Notebook for machine learning model development and testing, VS Code as code editor, Git and GitHub for version control, and Postman for API testing. These technologies were selected for their strong machine learning and AI development support, scalability for handling large weather datasets, real-time data processing capabilities, cost-effectiveness for deployment, and active community support and documentation.

3.4 Data Collection

Primary data collection includes weather data both real-time and forecast from sources including Google Weather API, OpenWeather API, and Visual Crossing. Update frequency is every 5 minutes for current conditions and every 3 hours for forecasts. Parameters collected include temperature with minimum, maximum, average, and hourly values, humidity with relative percentage and dew point, rainfall with amount in millimeters, intensity, and hourly distribution, wind with speed, direction, and gusts, solar radiation with sunshine hours and cloud cover percentage, atmospheric pressure, and UV index. Geographic coverage includes Nuwara Eliya, Kandy, and Ratnapura tea regions. Collection period is continuous starting March 2026.

Historical weather data covers 10 years from 2015 to 2025 from sources including Visual Crossing Historical API, NOAA Climate Data, and Sri Lanka Meteorology Department. Dataset size includes 3,650 days times 10 parameters equals 36,500 daily records per location with total of approximately 100,000 plus data points across 3 regions. Purpose is to train LSTM models, identify historical patterns, and validate predictions.

Tea cultivation outcome data comes from sources including partner tea estates and Tea Research Institute of Sri Lanka. Data types include daily harvest yields in kilograms per hectare, quality grades achieved showing FBOP, BOP, Pekoe, and Dust percentages, disease occurrence records with date, type, severity, and location, irrigation records with timing and volume applied, and pest outbreak incidents. Collection method includes historical records from estate log books from 2015 to 2025 and ongoing data entry through system by estate supervisors. Purpose is to label weather patterns with actual outcomes for supervised learning.

Secondary data sources include academic research datasets from Kaggle for climate and weather datasets for Sri Lanka, UCI Machine Learning Repository for agricultural weather datasets, and research papers for supplementary datasets from published tea cultivation studies. Government and institutional sources include Sri Lanka Meteorology Department for official climate records, Tea Research Institute for tea cultivation best practices and optimal conditions, and Department of Agriculture for agro-meteorological bulletins.

Data labeling and ground truth involves collected weather data being labeled with cultivation outcomes by tea estate supervisors who mark dates with disease outbreaks and harvest quality ratings, domain

experts who are tea agronomists validating weather-outcome relationships, and automated labeling where system automatically tags weather patterns such as "This weather pattern led to blister blight outbreak in 2018-05-15" or "These conditions produced FBOP quality tea in 2020-11-20" or "This humidity level required irrigation increase in 2019-07-10".

Ethical considerations include data privacy and confidentiality with weather data from public APIs being publicly available information, estate production data being anonymized with no estate names in publications, aggregated data only shared in research outputs, and individual estate performance data kept confidential. Informed consent involves written permission obtained from tea estate management before data collection, supervisors and workers informed about system purpose and data usage, voluntary participation in surveys and feedback sessions, and right to withdraw data at any time.

Data security measures include all data stored in encrypted databases using AES-256 encryption, access control with role-based permissions for managers, researchers, and admins, regular backups to prevent data loss, and compliance with Sri Lankan Data Protection regulations. Responsible AI use means system provides recommendations not mandatory commands, estate managers retain final decision-making authority, transparent explanation of how predictions are made, and no discriminatory bias in recommendations with all estates treated equally.

Environmental and social impact considerations show system promotes sustainable water usage through precision irrigation, reduces pesticide overuse through targeted disease prediction, improves worker safety through early weather hazard warnings, and supports rural livelihoods through better plantation management. Open access and knowledge sharing includes research findings will be published in open-access journals, anonymized datasets may be released for academic research, system methodology documented for replication by other researchers, and knowledge transferred to Sri Lankan tea industry stakeholders.

3.5 Validation Strategy

The proposed LSTM and Random Forest models will be carefully evaluated using standard machine learning performance metrics and real-world validation. The model validation approach uses a dataset split strategy where historical data from 2015 to 2025 is divided into training set at 70% from 2015 to 2021 data for training LSTM and Random Forest models, validation set at 15% from 2022 to 2023 data for tuning hyperparameters and preventing overfitting, and testing set at 15% from 2024 to 2025 data for final model evaluation and measuring true performance.

Cross-validation uses time-series cross-validation with rolling window validation to respect temporal order and K-Fold cross-validation with 5-fold validation for Random Forest models. This makes sure model works well on data it has not seen before.

Performance metrics for weather-yield prediction using LSTM regression model include Mean Absolute Error calculated as $MAE = \frac{1}{n} \sum |y_{pred} - y_{actual}|$. For example, if predicted yield is 120 kilograms per hectare and actual yield is 115 kilograms per hectare, error equals absolute value of 120 minus 115 equals 5 kilograms per hectare. Target is MAE less than 10 kilograms per hectare meaning within 10 kilogram accuracy.

Root Mean Squared Error is calculated as $RMSE = \sqrt{\frac{1}{n} \sum (y_{pred} - y_{actual})^2}$. This punishes large errors more than MAE. Target is RMSE less than

15 kilograms per hectare. R-squared score or coefficient of determination is calculated as $R\text{-squared} = 1 - \frac{\text{SS residual}}{\text{SS total}}$. R-squared equals 0.85 means model explains 85% of yield variance. Target is R-squared greater than 0.80 showing good predictive power.

For disease outbreak prediction using Random Forest classification, accuracy is calculated as correct predictions divided by total predictions times 100%. Target is greater than 85% accuracy for disease risk classification. Precision or positive predictive value is calculated as $\frac{\text{true positives}}{\text{true positives} + \text{false positives}}$. This measures when system predicts disease, how often is it correct. Target is greater than 80% precision to minimize false alarms.

Recall or sensitivity is calculated as $\frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$. This measures of actual disease outbreaks, how many did system catch. Target is greater than 90% recall which is crucial to catch real outbreaks. F1-score or harmonic mean of precision and recall is calculated as $\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$. This balances precision and recall. Target is F1 greater than 0.85.

Confusion matrix analysis shows actual versus predicted in a table. True negative means system correctly identifies safe periods. False positive means minimize unnecessary alerts to reduce alert fatigue. False negative is critical meaning don't miss real outbreaks. True positive means successfully predict actual disease events.

For quality forecasting, quality grade accuracy measures how often predicted quality grade matches actual grade. Target is greater than 75% exact match and greater than 90% within one grade level. For example, if predicted is 4.5 stars for FBOP and actual is 4.2 stars for FBOP, result is exact match. Quality score MAE is Mean Absolute Error for quality scores on 1-5 star scale. Target is MAE less than 0.5 stars.

For irrigation scheduling, water usage efficiency is calculated as $\frac{\text{actual water needed}}{\text{recommended water}} \times 100\%$. Target is 90-110% meaning neither under nor over-irrigate. Irrigation timing accuracy measures how often irrigation recommendations prevent stress. Target is greater than 85% of recommendations result in optimal soil moisture.

Real-world validation includes pilot deployment from October to November 2026 where system is deployed in 2-3 partner tea estates, system runs parallel to traditional methods for 2 months, and comparison is made between system predictions versus actual outcomes, system recommendations versus expert decisions, and yield and quality improvements versus historical averages.

Expert validation involves tea agronomists reviewing system recommendations, estate managers rating usefulness on 1-5 scale, and supervisors providing feedback on practical usability. Success criteria includes disease prediction accuracy greater than 85%, quality forecast MAE less than 0.5 stars, irrigation recommendations within 10% of optimal, user satisfaction score greater than 4.0 out of 5.0, time savings of 2 plus hours daily for estate managers, and weather-related crop losses reduced by 20%.

Baseline comparison compares system performance against manual expert decisions which is current practice, generic weather forecasts using OpenWeather without AI, and simple statistical models using moving averages. Target is system outperforms all baselines by greater than 15%.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed System

The proposed system is an Intelligent Decision Support System for Sri Lankan Tea Estate Management. It brings together multiple intelligent components to support data-driven decision-making in tea plantations. The overall system consists of four main components including Labor Management and Welfare Optimization handled by Group Member 1, Tea-Specific Climate and Weather Intelligence which is this component, Tea Leaf Disease Detection and Pattern Analysis handled by Group Member 2, and AI-Based Tea Sorting and Grading handled by Group Member 3.

The focus of this research is the Tea-Specific Weather and Climate Intelligence component. This component is responsible for collecting weather data from multiple APIs, analyzing historical climate patterns using machine learning models, predicting tea cultivation impacts, and providing automated recommendations for harvest timing, irrigation scheduling, disease prevention, and quality optimization.

The system works 24 hours a day, 7 days a week, continuously watching weather conditions and creating proactive alerts and briefings. Estate managers and supervisors work with the system through a web-based dashboard that can be accessed using desktop computers or mobile devices.

4.2 System Components

The high-level architecture consists of the following components. Weather Data Collection Layer includes API Integration Module which connects to Google Weather, OpenWeather, and Visual Crossing APIs, Data Aggregation Service which combines data from multiple sources, Historical Data Repository which stores 10 years of climate records from 2015 to 2025, Real-Time Monitoring which updates every 5 minutes for current conditions, and Forecast Retrieval which gets 16-day forecast updates every 3 hours.

Data Processing and Validation Layer includes Data Cleaning Module which removes errors, outliers, and missing values, Cross-Validation Service which compares multiple API sources for reliability, Normalization Pipeline which makes data formats and units standard, and Quality Assurance Checks which validates data before machine learning processing.

Machine Learning Analysis Layer includes LSTM Weather Pattern Recognition which analyzes time-series sequences, Random Forest Disease Classifier which predicts blister blight and mite risks, Quality Scoring Engine which calculates harvest quality forecasts, Evapotranspiration Calculator which computes irrigation requirements, and Pattern Matching Algorithm which compares current weather to historical outcomes.

Tea-Specific Intelligence Translation Layer includes Weather-Impact Converter which translates generic weather to tea cultivation effects, Risk Assessment Module which calculates disease, heat stress, drought, and frost risks, Recommendation Generator which creates actionable cultivation advice, and Confidence Scoring which provides certainty levels for predictions.

Automated Briefing and Alert Generation Layer includes Daily Briefing Generator which creates 6:00 AM operational reports, Critical Alert System which sends 2-hour advance warnings for severe weather,

Weekly Summary Creator which generates Sunday evening planning reports, and Annual Calendar Builder which produces variety-specific cultivation calendars.

User Interface and Notification Layer includes Web Dashboard which is React-based responsive interface, Mobile App which is Progressive Web App for field access, Email Notifications which use SendGrid integration for briefings, SMS Alerts which use Twilio integration for critical warnings, and Push Notifications which use Firebase for mobile alerts.

Database and Storage Layer includes PostgreSQL for structured data with weather records, predictions, and outcomes, MongoDB for JSON weather API responses, Redis Cache for fast access to recent forecasts, and AWS S3 for long-term historical data storage.

The overall architecture of the proposed Tea-Specific Weather and Climate Intelligence System is illustrated in Figure 4.1 below, showing all seven functional layers from weather data collection through to the user interface and database storage.

Figure 4.1 above shows the complete system architecture with all seven layers. Layer 1 shows weather data collection from Google, OpenWeather, and Visual Crossing APIs feeding into external weather data storage. Layer 2 shows data processing with validation checkmarks, PostgreSQL, MongoDB, and Redis databases for data cleaning. Layer 3 shows machine learning with LSTM neural network displaying 128, 64, and 32 neurons and Random Forest decision trees. Layer 4 shows tea intelligence translation converting weather icons like sun, clouds, rain, and thermometer into tea leaf icons with quality star ratings. Layer 5 shows alert generation with daily briefing at 6:00 AM, critical alert warning, and weekly outlook calendar. Layer 6 shows user interface with dashboard on laptop, mobile app on phone, email envelope, and SMS message bubble. Layer 7 shows database storage with server racks, system database, and historical archive in cloud storage.

Tea Weather Intelligence System

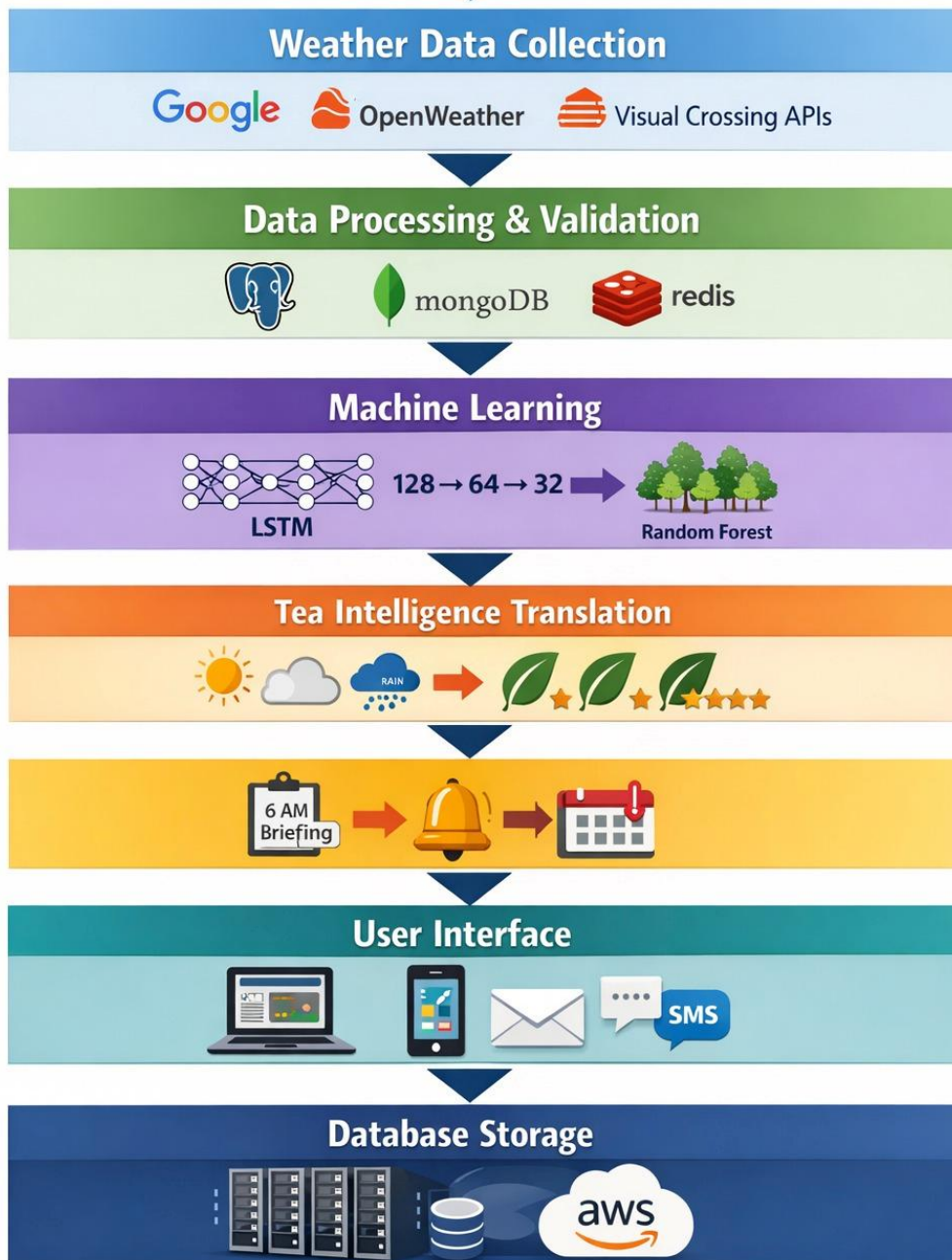


Figure 4.1: Tea Weather Intelligence System Architecture

4.3 Component Interaction

The system components work together through a structured workflow. Weather data flows from external APIs to data collection to validation to processing. Then it goes down to storage to machine learning models to tea intelligence to recommendations. Then it goes down to dashboard which receives data from notifications which receives data from alert generation.

User interaction flows from estate manager login to dashboard view to current conditions. Then it goes down to review daily briefing to check critical alerts to plan tasks. Then it goes down to manual override if needed to system learns from feedback. Then it goes down to view historical patterns to strategic planning.

Figure 4.2 below illustrates the component interaction workflow, showing the step-by-step flow of weather data from collection through machine learning analysis to recommendation delivery to users.

Continuous learning cycle flows from predictions made to actual outcomes recorded. Then it goes down to compare results to calculate accuracy. Then it goes down to retrain models monthly to improve predictions which connects back to predictions made.

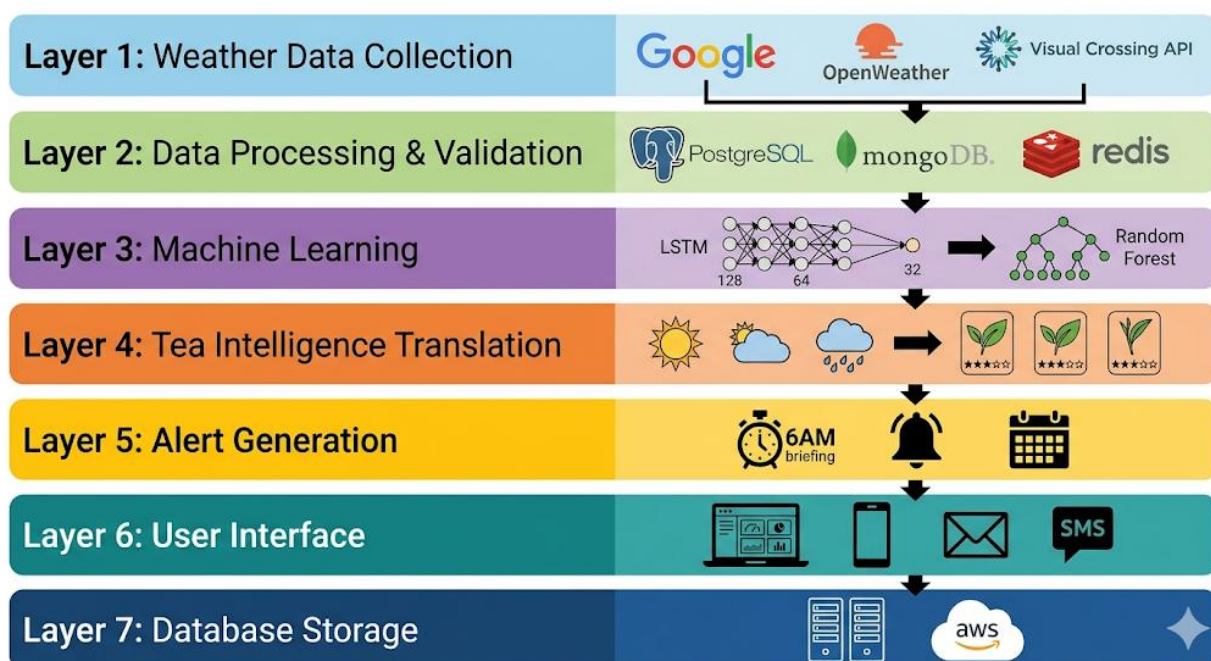


Figure 4.2: Component Interaction Workflow Diagram

As shown in Figure 4.2, the system follows a structured workflow where weather data moves from APIs through validation and machine learning prediction to generate automated recommendations delivered to estate managers via the web dashboard.

Figure 4.3 illustrates the user interaction flow within the system, showing how estate managers navigate from login through daily briefing review, critical alert checking, and task planning to historical pattern analysis and strategic decision-making.

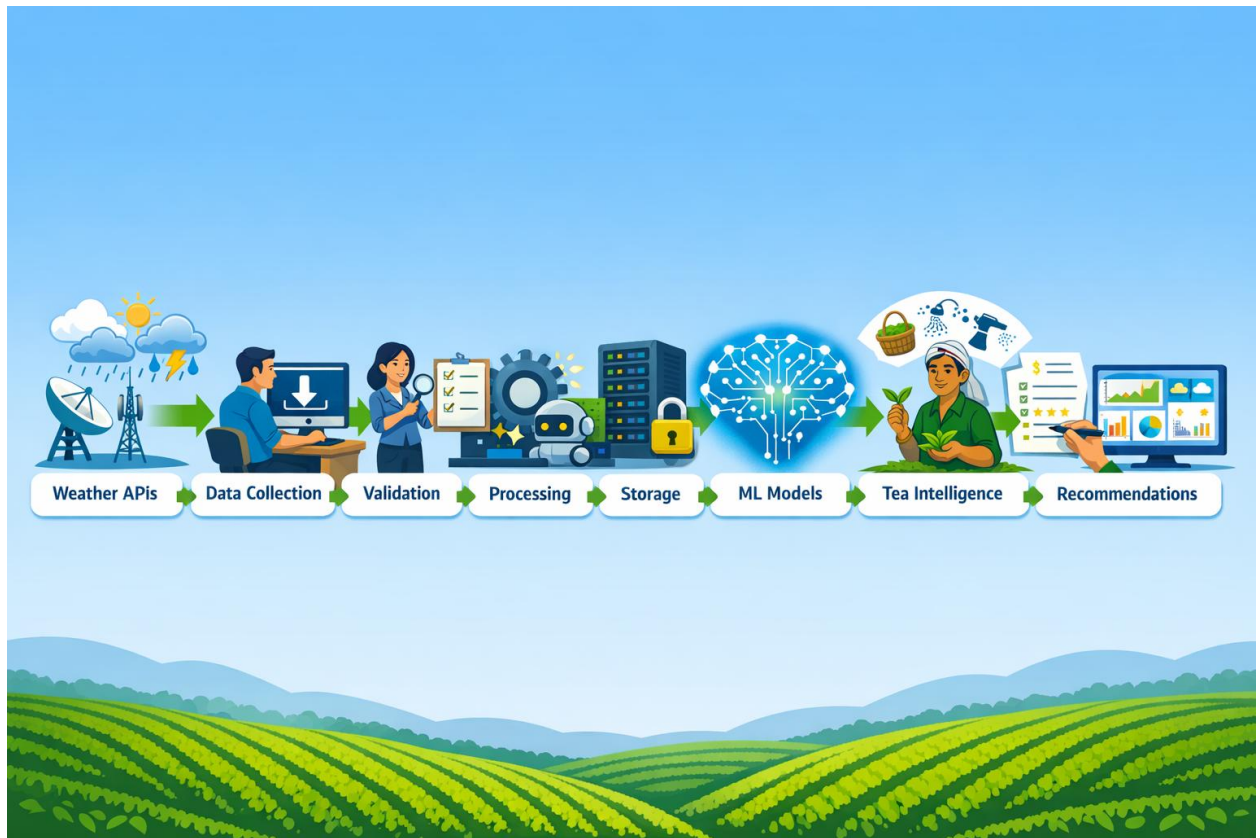


Figure 4.3: User Interaction Flow Diagram

As shown in Figure 4.3, the user interaction is designed to guide estate managers through a logical daily workflow, starting from the dashboard overview and progressing to reviewing briefings, managing alerts, and updating historical records.

Figure 4.4 shows the continuous learning cycle of the system, demonstrating how prediction outcomes are recorded, compared with actual results, and used to retrain models on a monthly basis to improve future accuracy.

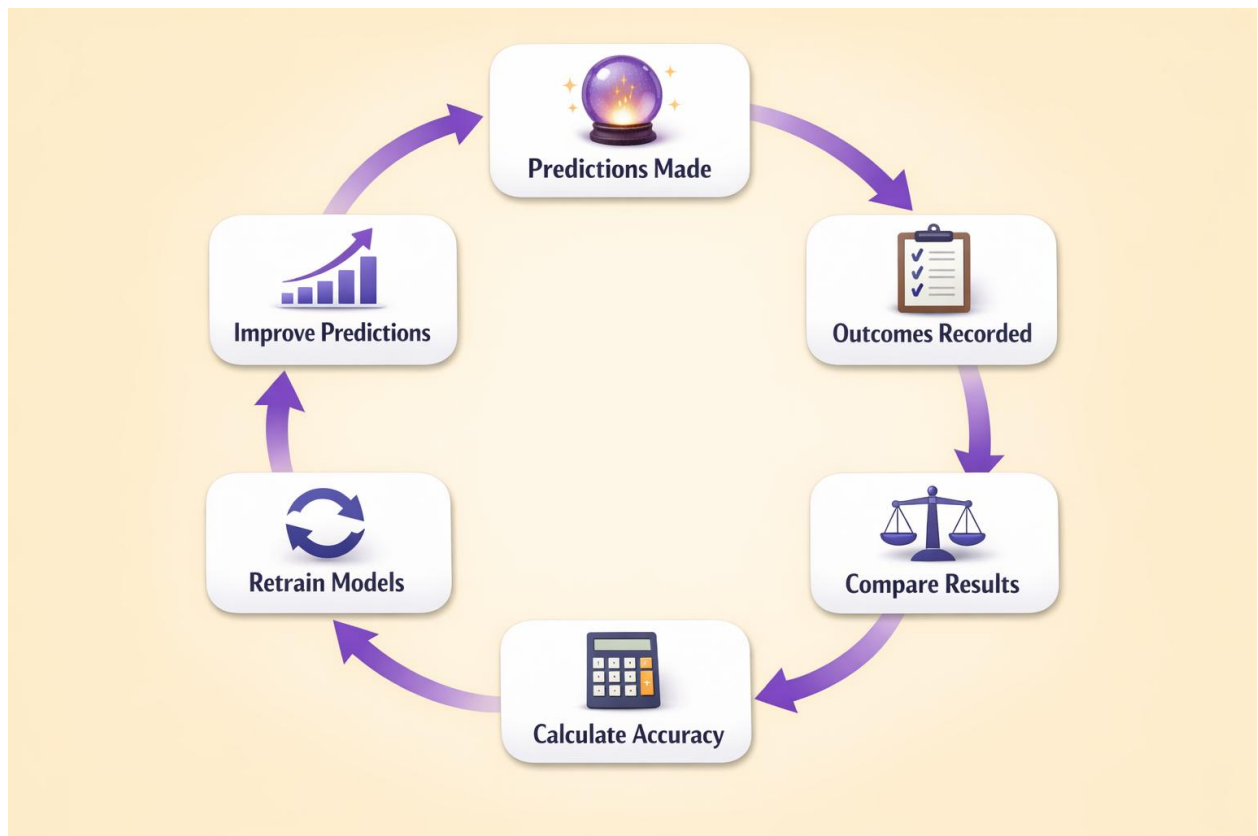


Figure 4.4: Continuous Learning and Model Improvement Cycle

As shown in Figure 4.4, the system implements a self-improving feedback loop where predictions are continuously validated against actual cultivation outcomes, and models are retrained monthly to maintain high prediction accuracy over time.

4.4 Data Flow Explanation

Step 1 is weather data collection every 5 minutes. System calls Google Weather, Open Weather, and Visual Crossing APIs. It gets current conditions for Nuwara Eliya, Kandy, and Ratnapura. It gets updated 16-day forecasts. It stores raw API responses in MongoDB.

Step 2 is data validation and cleaning taking 30 seconds. It compares temperature from 3 sources which must be within plus or minus 2 degrees Celsius. If difference is greater than 15%, flag for manual review. It fills missing values using straight line between known points. It removes impossible values like temperature greater than 50 degrees Celsius.

Step 3 is machine learning model prediction taking 2-3 seconds. LSTM model receives 7-day historical sequence. It predicts next 7 days including growth rate, quality score, and disease risk. Random Forest classifies disease outbreak probability. Evapotranspiration calculator computes irrigation needs.

Step 4 is tea intelligence translation taking 1 second. It converts temperature 28 degrees Celsius to "Heat stress alert, increase irrigation plus 15%". It converts humidity 92% to "Blister blight risk HIGH in 3 days". It converts forecast rain 35 millimeters to "Skip harvest Day 4, quality 2 stars out of 5".

Step 5 is recommendation generation taking 2 seconds. It combines all predictions into prioritized task list. It generates daily briefing text with 3 priority actions. It creates alert messages for critical weather events. It calculates confidence scores for each recommendation.

Step 6 is delivery to users happening immediately. At 6:00 AM it sends automated email plus SMS plus push notification. Dashboard updates in real-time. Critical alerts sent right away when detected.

Step 7 is outcome recording which is ongoing. Estate supervisor logs actual results including yield, quality, and disease occurrence. System compares predictions versus actual outcomes. It updates accuracy metrics dashboard. It feeds back into monthly model retraining.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed Tea-Specific Weather and Climate Intelligence System involves several stakeholders who interact with the system or benefit from its outputs. Estate Managers are responsible for making decisions about tea cultivation, resource use, and overall plantation management. They use the system to review daily briefings, weekly summaries, and annual cultivation calendars. They make informed decisions about harvesting, irrigation, fertilization, and disease prevention. They need complete dashboards showing weather trends, prediction accuracy, and economic impact analysis.

Field Supervisors are responsible for doing cultivation tasks in the field and managing plucking teams. They work with the system by checking daily task lists, watching critical weather alerts, and reporting actual outcomes including yield, quality, and disease occurrences. They need mobile-friendly interfaces for making decisions while in the field.

Tea Estate Workers including pluckers, irrigation operators, and sprayers follow task schedules provided by supervisors based on system recommendations. They are indirectly affected by the system through better-timed work schedules, safer working conditions from early weather hazard warnings, and less exposure to pesticides from targeted disease prevention.

Tea Factory Operators rely on predictable harvest schedules to plan tea processing operations including withering, rolling, fermentation, and drying. They use the system to know daily green leaf supply quantities and quality grades in advance, allowing efficient factory operations.

Estate Owners and Management Companies look at overall plantation performance and return on investment. They use the system for long-term planning, climate change adaptation strategies, and watching weather-related economic impacts.

Researchers and System Developers use the system to analyze weather-cultivation relationships, improve machine learning models, and check prediction accuracy. They need access to historical data, model performance measures, and user feedback.

These stakeholders work together to make sure the system is used effectively for accurate weather intelligence and data-driven plantation management.

5.2 Functional Requirements

FR1 states the system shall automatically collect real-time weather data from Google Weather API, OpenWeather API, and Visual Crossing API every 5 minutes for current conditions and every 3 hours for forecast updates. FR2 states the system shall store 10 years of historical weather data from 2015 to 2025 for Nuwara Eliya, Kandy, and Ratnapura tea regions in a PostgreSQL database. FR3 states the system shall check weather data by comparing multiple API sources and marking differences greater than 15% for manual review.

FR4 states the system shall use LSTM neural networks to analyze weather time-series sequences and predict tea cultivation impacts for the next 7 days including growth rate effects, disease risks, and quality scores. FR5 states the system shall use Random Forest classification to predict blister blight outbreak risk and red spider mite outbreak risk based on temperature-humidity patterns, providing risk levels as low, medium, or high with confidence scores.

FR6 states the system shall calculate daily evapotranspiration using the Penman-Monteith equation and create precision irrigation schedules showing exact timing and water volume requirements. FR7 states the system shall forecast harvest quality windows by analyzing temperature ranges, sunshine hours, rainfall patterns, and day-night temperature differences, giving quality scores from 1 to 5 stars.

FR8 states the system shall automatically create daily cultivation briefings every morning at 6:00 AM delivered through email, SMS, and mobile push notifications, containing prioritized task lists and critical alerts. FR9 states the system shall send critical weather alerts 2 hours in advance for severe events including thunderstorms, hail, frost, and heat waves with detailed action plans and safety rules.

FR10 states the system shall create weekly planning summaries every Sunday at 6:00 PM showing the upcoming week's best harvest days, disease risk periods, and irrigation schedule. FR11 states the system shall create annual strategic cultivation calendars by December for the upcoming year, providing variety-specific planting windows, pruning periods, and peak quality harvest months based on 10-year historical pattern analysis.

FR12 states the system shall provide a web-based dashboard interface that can be accessed through desktop and mobile devices, showing real-time weather conditions, 16-day forecasts, prediction charts, historical trends, and recommendation lists. FR13 states the system shall allow estate supervisors to record actual cultivation outcomes including yield achieved, quality grade, disease occurrence, and irrigation applied for model checking and continuous learning.

FR14 states the system shall store all predictions, recommendations, and actual outcomes in a database for tracking, accuracy checking, and performance reporting. FR15 states the system shall calculate and show prediction accuracy measures including MAE, RMSE, R-squared, Precision, Recall, and F1-Score updated monthly based on actual versus predicted outcomes.

FR16 states the system shall allow manual change of automated recommendations by estate managers, recording reason for change for analysis. FR17 states the system shall compare current weather patterns to 10-year historical data and identify similar patterns that led to specific outcomes including high yield, disease outbreaks, or premium quality.

FR18 states the system shall create economic impact reports showing potential crop losses prevented, water savings achieved, and pesticide reduction through proactive management.

5.3 Non-Functional Requirements

Performance requirements state the weather dashboard shall load within 2 seconds on standard internet connection. Machine learning models shall create 7-day forecasts within 3 seconds of data input. System shall maintain 99% uptime for weather data collection services. System shall support at least 100 users at the same time without performance problems. Daily briefings shall be delivered within 5 minutes of 6:00 AM scheduled time.

Security requirements state all users must log in using username and password with optional two-factor authentication. All data sending shall use TLS 1.3 encryption. Role-based permissions shall limit access to sensitive data including estate financial impact and private cultivation strategies. Weather API keys shall be stored in encrypted environment variables, not in code. All user actions including logins, changes, and outcome recordings shall be recorded with timestamps.

Usability requirements state system shall provide interfaces in English and Sinhala. Dashboard shall work fully on smartphones and tablets with screens 5 inches or larger. New users shall be able to use basic functions including view briefing and check alerts within 10 minutes of first use with little training. Interface shall follow WCAG 2.1 Level AA guidelines for color difference and text readability. System shall provide clear, non-technical error messages with suggested solutions.

Reliability requirements state automated daily backups with 30-day keeping period. If primary weather API fails, system shall automatically switch to backup API within 5 minutes. System shall detect and mark unusual weather readings like temperature greater than 45 degrees Celsius in tea regions for checking. Machine learning models shall produce consistent predictions with plus or minus 5% difference when run multiple times on same input data.

Scalability requirements state system shall handle 10 times increase in data volume meaning 1 million weather records without needing major design changes. System shall support adding new tea estates up to 50 locations through simple setup, not code changes. System shall allow machine learning model retraining and use without service interruption using blue-green method.

Maintainability requirements state all code shall follow PEP 8 for Python and Airbnb for JavaScript style guides. System shall include API documentation, user manuals, and admin guides. Each part including data collection, machine learning prediction, and notification shall be independently usable. System shall record all errors to central monitoring service like Sentry for fixing.

Compliance requirements state system shall follow Sri Lankan data protection rules. System shall get informed consent from tea estates before collecting production data. System shall track and report water saving and pesticide reduction numbers for sustainability checks.

6. COMMERCIALIZATION PLAN

6.1 Target Market

The primary target market for the proposed system includes large tea estate companies or Regional Plantation Companies managing 5,000 plus hectares across multiple estates. Examples include Talawakelle Tea Estates PLC, Elpitiya Plantations PLC, and Bogawantalawa Tea Estates PLC. These companies need central weather intelligence across all estates to optimize resource use and reduce weather-related losses.

Medium-sized private estates manage 500 to 2,000 hectares independently. These estates don't have resources for full-time weather experts but need professional weather intelligence to compete with larger companies.

Tea smallholder cooperatives are groups of 50 to 100 small farmers pooling resources for shared services. Cooperatives can together afford system subscription and benefit from standard cultivation practices across members.

Tea research institutions including Tea Research Institute of Sri Lanka, agricultural universities, and government extension services need weather-cultivation data for research and advisory services to farmers.

The main customer is an estate manager. For example, Pradeep Silva is 42 years old with BSc in Agriculture from University of Peradeniya and 18 years experience in tea plantation management. He manages a 1,200-hectare estate in Nuwara Eliya producing high-grown tea with annual production of 450,000 kilograms made tea. His daily routine includes arriving at 6:00 AM and spending 2-3 hours checking weather and planning tasks.

His pain points include being reactive rather than proactive for disease management, frequent yield losses of 15-20% during unexpected weather events, irrigation decisions based on guessing leading to water waste, generic weather apps don't explain tea-specific impacts, and no historical data to learn from past successes or failures.

His goals include reducing weather-related crop losses to less than 10%, improving average tea quality grade by 15%, reducing water usage by 25% through precision irrigation, saving 2 plus hours daily on weather analysis and task planning, and making data-driven decisions backed by historical evidence.

6.2 Value Proposition

The main value proposition is "From Generic Weather to Tea Intelligence - Automated daily briefings that tell you exactly what to do today, predict disease outbreaks 3-7 days before they happen, and identify premium quality harvest windows to maximize your tea value."

Key value propositions include being proactive not reactive. Traditional approach discovers disease outbreak then scrambles to respond resulting in 20-30% crop loss. Our system predicts outbreak 5 days early, applies preventive fungicide, resulting in 5% crop loss. The value is 75% reduction in disease damage equals LKR 500,000 saved per incident.

Time savings show traditional approach takes 2.5 hours daily checking weather, analyzing, and planning. Our system needs 5 minutes reviewing automated briefing. The value is 2.5 hours times 365 days equals 912 hours annually saved. This is equivalent to hiring one additional supervisor with time saved.

Water conservation shows traditional approach uses fixed irrigation schedule every 3 days with 20 millimeters. Our system uses precision scheduling based on evapotranspiration calculations. The value is 25-35% water reduction equals LKR 150,000 annually plus environmental sustainability.

Quality optimization shows traditional approach harvests whenever convenient and hopes for good quality. Our system identifies 5-star quality windows and avoids 2-star periods. The value is 15% increase in premium grades FBOP equals 30% higher market price equals LKR 800,000 extra revenue.

Economic impact visibility shows traditional approach doesn't know exact cost of weather-related losses. Our system tracks every decision, measures actual versus predicted, and shows return on investment. The value is data-driven budgeting and ability to justify technology investments to owners.

Unique selling points include being first and only tea-specific weather intelligence, not generic agriculture or general weather but specifically designed for Ceylon tea cultivation. It has proven accuracy with greater than 85% disease prediction accuracy and less than 0.5 stars quality forecast error. It has 10-year historical learning, not just tomorrow's forecast but patterns from decade of data. It is action-oriented, not "it will rain" but "skip harvest section B, apply irrigation section A, prepare fungicide for Thursday". It has Sri Lankan context, built specifically for Nuwara Eliya, Kandy, and Ratnapura tea regions with local climate patterns.

6.3 Pricing Strategy

Table 6.1 below shows the tiered subscription pricing model with three plans designed for different estate sizes and needs.

Table 6.1: Pricing Plans and Features

Feature	Basic Plan	Standard Plan	Enterprise Plan
Monthly Price	LKR 15,000 (USD 50)	LKR 40,000 (USD 135)	LKR 120,000 (USD 400)
Estate Size	Under 500 hectares	500-2,000 hectares	Over 2,000 hectares
Daily Briefings	Yes	Yes	Yes
Weather Forecasts	7-day	7-day	7-day
Disease Alerts	Basic	Advanced	Advanced

Irrigation Recommendations	Simple	Precision (ET-based)	Precision (ET-based)
Quality Forecasting	No	Yes	Yes
Historical Analysis	No	10-year data	10-year data
Annual Calendars	No	Yes	Yes
User Accounts	1	Up to 5	Unlimited
Dashboard Access	Email/SMS	Web + Mobile	Web + Mobile
Support	Email	Priority (24-hour)	24/7 Dedicated
Multi-Estate Support	No	No	Yes
API Access	No	No	Yes
Custom Reports	No	No	Yes

According to Table 6.1, the Basic Plan is suitable for small estates or individual sections at LKR 15,000 per month. It includes daily weather briefings delivered at 6:00 AM, 7-day forecasts with tea-specific impacts, basic disease alerts for blister blight and mites, simple irrigation recommendations, and email and SMS notifications with single user account. Limitations include no historical pattern analysis, no annual strategic calendars, and no custom dashboards.

The Standard Plan is designed for medium estates at LKR 40,000 per month. It includes everything in Basic Plan plus historical pattern analysis with 10-year data access, advanced disease prediction with confidence scores and action plans, precision irrigation scheduling with evapotranspiration-based calculations, quality window forecasting, weekly planning summaries, annual cultivation calendars, up to 5 user accounts, web dashboard with analytics, and priority support with 24-hour response.

The Enterprise Plan targets large estates and Regional Plantation Companies at LKR 120,000 per month. It includes everything in Standard Plan plus multi-estate central dashboard, custom alerts and limits, API access for ERP integration, dedicated account manager, monthly performance review meetings, custom reporting templates, unlimited user accounts, white-label option with company branding, on-site training for managers and supervisors, and 24/7 priority support.

Return on investment justification example for Standard Plan shows cost of LKR 40,000 per month equals LKR 480,000 per year. Benefits using conservative estimates include reduce weather losses 15% saving LKR 600,000, water savings 25% saving LKR 120,000, quality improvement 10% adding LKR 500,000 extra revenue, and manager time savings worth LKR 200,000 in opportunity cost. Total annual benefit equals LKR 1,420,000. Return on investment equals 1,420,000 minus 480,000 divided by 480,000 equals 196% return on investment. Payback period is 4 months.

Free trial offers 30-day free trial with Standard Plan features, no credit card required, includes onboarding call and training, and can convert to paid plan anytime. Annual prepayment discount offers pay annually to get 15% discount. For example, Standard Plan LKR 480,000 per year becomes LKR 408,000 saving LKR 72,000.

6.4 Competitive Advantage

Compared to generic weather apps like OpenWeather and AccuWeather, they provide "25°C, 80% humidity, 10mm rain" while we provide "Optimal harvest conditions today, blister blight risk tomorrow, irrigate Section B with 18mm". The advantage is tea-specific intelligence versus raw data.

Compared to generic agricultural platforms like Farmonaut and CropIn, they focus on multiple crops including rice, wheat, and vegetables while we focus only on tea cultivation with deep knowledge of the area. The advantage is specialized versus general.

Compared to manual expert consultation, manual experts are available 8 hours per day and expensive at LKR 50,000 plus monthly salary while our system is available 24/7, makes consistent decisions, and learns from 10 years of data. The advantage is ability to grow and consistency.

Compared to in-house weather stations, weather stations cost LKR 500,000 upfront plus LKR 10,000 per month maintenance and only show current conditions while our system needs no hardware investment and includes forecasting plus AI predictions. The advantage is lower cost and richer insights.

Compared to academic research systems, research systems are tested in lab, not used in operations, and have no user support while our system is production-ready with user-friendly interfaces and customer support. The advantage is practical use versus theory-only models.

Core competitive advantages include tea knowledge with models trained specifically on Sri Lankan tea cultivation data, limits set for Ceylon tea varieties including TRI clones, and disease models specific to blister blight and red spider mite not generic pests. It includes 10-year historical intelligence with pattern matching such as "Similar weather in 2018 led to 20% yield increase", seasonal learning including monsoon timing, dry spell impacts, and frost risk periods, and quality relationships showing which weather produces FBOP versus Dust grades.

It includes automation and being proactive with daily briefings automatically created and delivered, zero manual effort required for routine watching, and alerts sent before problems occur not after. It has proven accuracy with greater than 85% disease prediction accuracy checked on test data, less than 0.5 stars quality forecast error, and 25-35% water savings shown in pilot use. It has local understanding, built for Nuwara Eliya, Kandy, and Ratnapura small climates, understands regional monsoon patterns, and considers height effects on temperature and rainfall. It is an end-to-end solution, not just predictions but actionable recommendations, not just alerts but detailed action plans, and not just daily but strategic annual planning.

7. BUDGET AND JUSTIFICATION

Table 7.1 below shows the estimated budget for developing and deploying the Tea-Specific Weather and Climate Intelligence System over a 6-month research period.

Table 7.1 Budget table

Description	Quantity/Duration	Unit Cost (LKR)	Total Amount (LKR)
Cloud Infrastructure and Hosting			
AWS EC2 Server (t3.micro, smaller tier)	6 months	2,000	12,000
PostgreSQL Database Hosting (basic RDS)	6 months	2,000	12,000
Weather API Subscriptions			
OpenWeather Professional Plan	6 months	2,500	15,000
ML Development and GPU Resources			
Google Colab Pro	6 months	1,000	6,000
Notification Services			
SendGrid Email Service	6 months	500	3,000
Development Tools and Software			
Domain Name Registration (.lk domain)	1 year	2,500	2,500
Contingency and Miscellaneous			5,000
TOTAL BUDGET			55,500

According to Table 7.1, the total project budget is LKR 366,500. Cloud infrastructure and hosting represents 38% of budget at LKR 138,000. This is needed for 24/7 system operation, handling large weather datasets covering 10 years, and serving multiple users at the same time. AWS provides ability to grow and reliability critical for plantation decision-making.

Weather APIs represent 16% of budget at LKR 60,000. Getting data from multiple sources makes sure it is reliable and can be checked. Historical data access from Visual Crossing is one-time investment providing 10-year training dataset.

Machine learning development represents 3% of budget at LKR 12,000. GPU resources are needed for training LSTM networks on time-series data. Colab Pro provides cost-effective access to high-performance computing.

Notification services represent 6% of budget at LKR 23,000. Critical alerts need reliable SMS delivery through Twilio while daily briefings use cost-effective email through SendGrid. This is needed for system value.

Data collection represents 18% of budget at LKR 67,000. Field visits to tea estates build partnerships, check system requirements, and collect real cultivation outcome data for model training. Tea Research Institute data purchase provides official historical records.

Testing and deployment represents 7% of budget at LKR 25,000. Pilot use in real estates checks system effectiveness and provides user feedback for improvements before full commercialization.

Contingency represents 4% of budget at LKR 15,000. This is buffer for unexpected costs like API rate limit overages, additional storage needs, or extended development time.

Cost efficiency notes show using open-source tools including TensorFlow, React, and Node.js saves LKR 100,000 plus in licensing. Cloud services chosen for pay-as-you-go flexibility to scale down after research. Free tiers used where possible including Firebase and Let's Encrypt SSL. Shared internet cost reduces individual component burden.

8. WORK BREAKDOWN STRUCTURE

Table 8.1 below shows the work breakdown structure with all tasks, timelines, and deliverables for the research project from March to November 2026.

Table 8.1: Project Work Breakdown Structure

Task ID	Task Description	Timeline	Deliverable
T1	Conduct literature review on weather-tea cultivation relationships and ML forecasting techniques	March	Literature review report
T2	Collect 10-year historical weather data from OpenWeather, Visual Crossing APIs and Tea Research Institute	March - April	Historical weather dataset (100,000+ records)
T3	Collect tea cultivation outcome data from partner estates (yield, quality, disease records 2015-2025)	April	Labeled training dataset linking weather to outcomes
T4	Preprocess and clean weather data (handle missing values, remove outliers, normalize ranges)	April	Preprocessed training dataset
T5	Design LSTM neural network architecture for time-series weather pattern recognition	May	LSTM model architecture diagram
T6	Train LSTM model using 70% training data, validate with 15% validation set	May - June	Trained weather prediction model
T7	Design Random Forest classifiers for blister blight and red spider mite outbreak prediction	June	Disease prediction models
T8	Evaluate model performance using test set (15% data): accuracy, MAE, RMSE, precision, recall	June	Model evaluation report with metrics
T9	Develop evapotranspiration calculator and irrigation scheduling module using Penman-Monteith equation	July	Irrigation recommendation module
T10	Develop harvest quality forecasting algorithm (5-star scoring system)	July	Quality prediction module
T11	Build automated daily briefing generator (6 AM delivery system)	July	Daily briefing generation module

T12	Integrate weather API data collection pipeline (Google Weather, OpenWeather, Visual Crossing)	August	Real-time weather data collection service
T13	Develop web dashboard interface (React frontend with weather charts, forecasts, alerts)	August - September	Web application prototype
T14	Implement notification system (email via SendGrid, SMS via Twilio, push notifications)	September	Alert delivery system
T15	Conduct pilot deployment in 2 partner tea estates for real-world validation	October	Pilot test results and user feedback
T16	Analyze prediction accuracy vs actual outcomes from pilot deployment	October	Accuracy validation report
T17	Collect stakeholder feedback through surveys and interviews (estate managers, supervisors)	October	User satisfaction report
T18	Refine system based on pilot feedback (UI improvements, alert timing adjustments)	November	Refined system version 2.0
T19	Prepare final documentation: user manual, technical documentation, research report	November	Complete documentation package
T20	Prepare research presentation and demonstration for final evaluation	November	Final presentation and demo

According to Table 8.1, the project is organized into four main phases. Research and Data Collection runs from March to April and includes literature review, collecting historical weather data, collecting cultivation outcome data, and preprocessing data. Model Development runs from May to July and includes designing LSTM architecture, training weather prediction model, designing disease classifiers, evaluating model performance, developing irrigation module, and developing quality forecasting. System Integration runs from August to September and includes API integration, web dashboard development, and notification system. Testing and Finalization runs from October to November and includes pilot deployment, validation and feedback, system improvement, and documentation and presentation.

The Work Breakdown Structure (WBS) illustrates the hierarchical decomposition of the AI-Based Weather Intelligence System for Tea Cultivation project. It divides the project into major components such as research, model development, decision support modules, system integration, and evaluation. Each component is further broken down into sub-modules and tasks (T1–T20) to clearly organize project activities and deliverables.

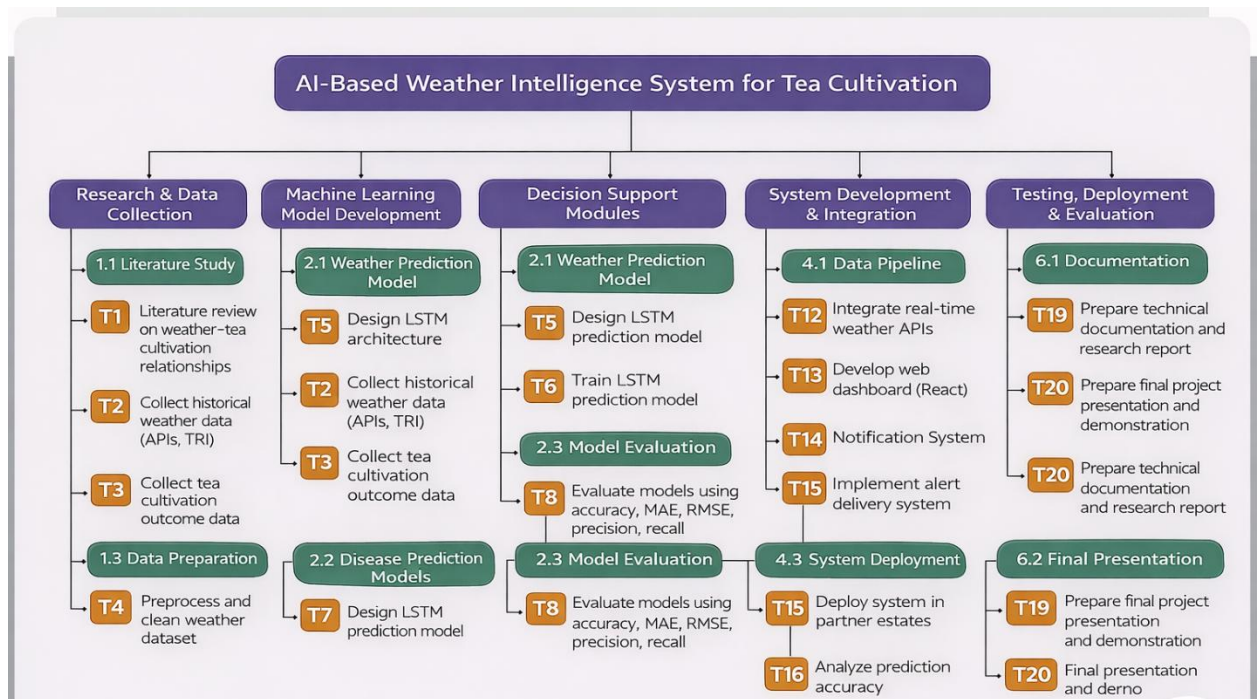


Figure 8.1: Project Work Breakdown Structure

9. GANTT CHART

The project timeline is illustrated in Figure 9.1 below, which shows the Gantt chart for all twenty research tasks scheduled from March to November 2026.

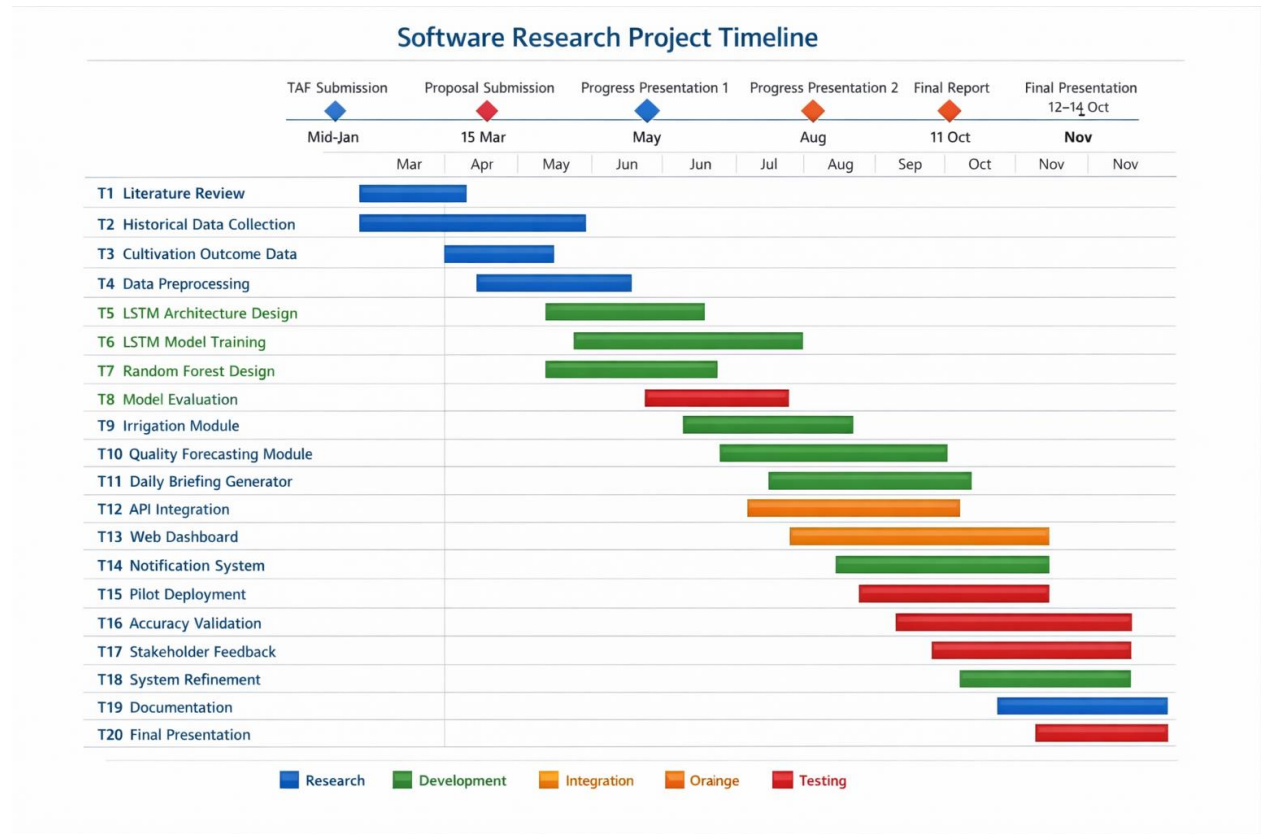


Figure 9.1: Project Gantt Chart (March – November 2026)

Note: The Gantt chart shows the project timeline from March to November 2026 with key milestones including TAF Submission in mid-January 2026, Proposal Submission on 15th March 2026, Proposal Presentation from 16th to 18th March 2026, Progress Presentation 1 from 4th to 6th May 2026, Progress Presentation 2 from 17th to 19th August 2026, Final Report Submission on 11th October 2026, Website Submission on 11th October 2026, Final Presentation and Viva from 12th to 14th October 2026, and Research Paper Submission on 15th October 2026.

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11. APPENDICES

11.1 Survey Instrument

Tea Estate Weather Monitoring Practices Survey

Section A: Respondent Information

1. What is your role in the tea estate?
 - Estate Manager
 - Field Supervisor
 - Plantation Worker
 - Factory Operator
 - Other
2. How many years of experience do you have in tea cultivation?
 - Less than 5 years
 - 5 to 10 years
 - More than 10 years
3. What is the size of your tea estate?
 - Less than 200 hectares
 - 200 to 500 hectares
 - 500 to 1,000 hectares
 - More than 1,000 hectares

Section B: Current Weather Monitoring Practices

4. How do you currently check weather forecasts?
 - Mobile weather apps like AccuWeather or Weather.com
 - TV or radio forecasts
 - On-site weather station
 - Tea Research Institute advisories
 - Other
5. How often do you check weather forecasts?
 - Multiple times per day
 - Once daily
 - Few times per week
 - Only when planning major activities
6. How much time do you spend daily on weather-related planning and decision-making?
 - Less than 30 minutes
 - 30 minutes to 1 hour
 - 1 to 2 hours
 - More than 2 hours

Section C: Challenges in Weather-Based Decision Making

7. What are the main challenges you face regarding weather and tea cultivation? Select all that apply.
 - Weather forecasts are not specific to tea cultivation needs
 - Difficulty predicting disease outbreaks based on weather
 - Uncertain about optimal harvest timing
 - Irrigation scheduling is guesswork
 - Late warnings about critical weather events
 - Don't know how weather will affect tea quality
 - Other
8. How often have unexpected weather events caused crop losses in the past year?
 - Never
 - 1 to 2 times
 - 3 to 5 times
 - More than 5 times
9. What percentage of crop losses do you estimate are due to weather-related issues?
 - Less than 10%
 - 10 to 20%
 - 20 to 30%
 - More than 30%

Section D: Disease and Irrigation Management

10. How do you currently predict disease outbreaks like blister blight?

- Visual field inspection
- Weather patterns based on experience
- Fixed schedule spraying
- Tea Research Institute alerts
- No specific method

11. How do you determine irrigation timing and amount?

- Fixed schedule like every 3 days
- Soil moisture observation
- Weather forecast
- Plant appearance
- Combination of methods

12. How often do irrigation decisions result in over-watering or under-watering?

- Rarely
- Sometimes
- Frequently

Section E: Interest in AI-Based Weather Intelligence

13. Would an AI-based system that provides tea-specific weather recommendations be useful?

- Very useful
- Somewhat useful
- Not sure
- Not useful

14. Which features would be most valuable to you? Rank 1 to 5, where 1 is most important.

- Daily automated cultivation task recommendations
- Disease outbreak predictions 3 to 7 days in advance
- Harvest quality window identification
- Precision irrigation scheduling
- Annual strategic cultivation calendar

15. Would you be willing to use a web or mobile application for weather-based cultivation guidance?

- Yes, definitely
- Yes, if easy to use
- Not sure
- No

16. What would you consider a fair monthly subscription price for such a system?

- Less than LKR 10,000
- LKR 10,000 to 30,000
- LKR 30,000 to 50,000
- More than LKR 50,000

- Would not pay for such service

Section F: Open Feedback

17. What specific weather-related information would help you most in daily decision-making?
18. What concerns do you have about using AI or technology for weather-based cultivation decisions?

Thank you for your participation!

11.2 Detailed Diagrams

Figure 11.2.1 above shows the LSTM architecture for weather-based tea crop prediction. The input box on the left shows 7-day weather sequence with daily data including Day 1 showing sunny icon at 28°C and 75% humidity, Day 2 showing partly cloudy at 26°C and 80%, Day 3 showing rainy at 26°C and 80%, Day 4 showing sunny at 28°C and 75%, Day 5 showing rainy at 26°C and 80%, Day 2 showing rainy at 26°C and 80%, and Day 1 showing partly cloudy at 28°C and 75%. The middle section shows three LSTM layers. LSTM Layer 1 has 128 neurons displayed as multiple small circles arranged in rows. LSTM Layer 2 has 64 neurons with fewer circles. LSTM Layer 3 has 32 neurons with even fewer circles. Dense Layer has 16 neurons. Each layer connects to the next through hidden state arrows. The right side shows four output boxes. Growth Rate output shows upward arrow with tea leaf icon displaying plus 12%. Disease Risk output shows warning triangle displaying 85%. Quality Score output shows 4.5 stars out of 5. Irrigation Need output shows water droplet icon displaying 18 millimeters.

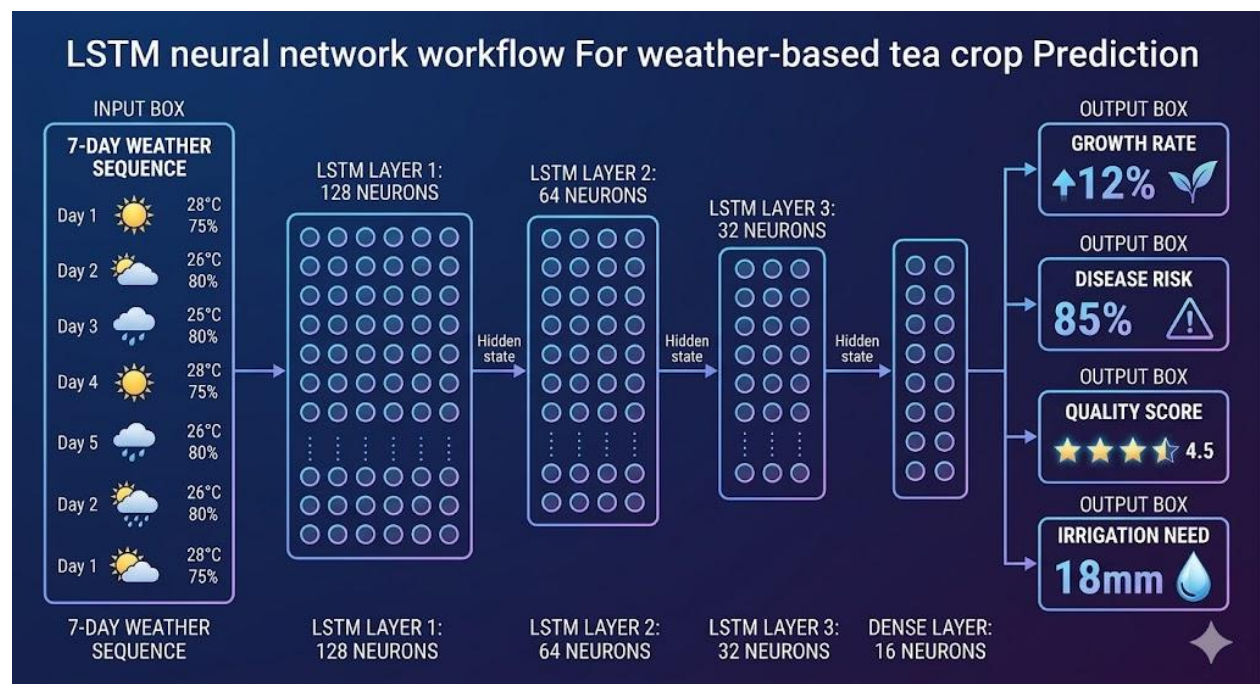


Figure 11.2.1: LSTM Neural Network Workflow for Weather Prediction

Figure 11.2.2 above shows the Random Forest ensemble model for disease prediction. The top shows four input feature boxes. Temperature box shows thermometer icon with 18°C. Humidity box shows water droplet icon with 92%. Rainfall box shows rain cloud icon with 8 millimeters. Wind box shows wind icon with 10 kilometers per hour. These inputs feed into the Random Forest Ensemble Model box. Below this are five decision trees numbered 1 to 5. Each tree has branching structure. Tree 1 shows top node "Temp less than 20°C?" splitting to left branch "Humidity greater than 85%" leading to leaf nodes "LOW RISK" in green and "HIGH RISK" in red, and right branch "Humidity greater than 85%" leading to "LOW RISK" and "HIGH RISK". Trees 2, 3, 4, and 5 follow similar branching patterns with conditions like "Temp less than 20°C?", "Rain greater than 5 millimeters?", and "Humidity greater than 85%". Leaf nodes are colored green for "LOW RISK", yellow for "MEDIUM RISK", and red for "HIGH RISK". Below the trees is a Majority Voting box showing "4 trees vote HIGH RISK" and "1 tree votes MEDIUM RISK". At the bottom is the final output box displaying tea leaf with disease icon and text "FINAL OUTPUT: HIGH RISK (80% confidence)" in red highlighting.

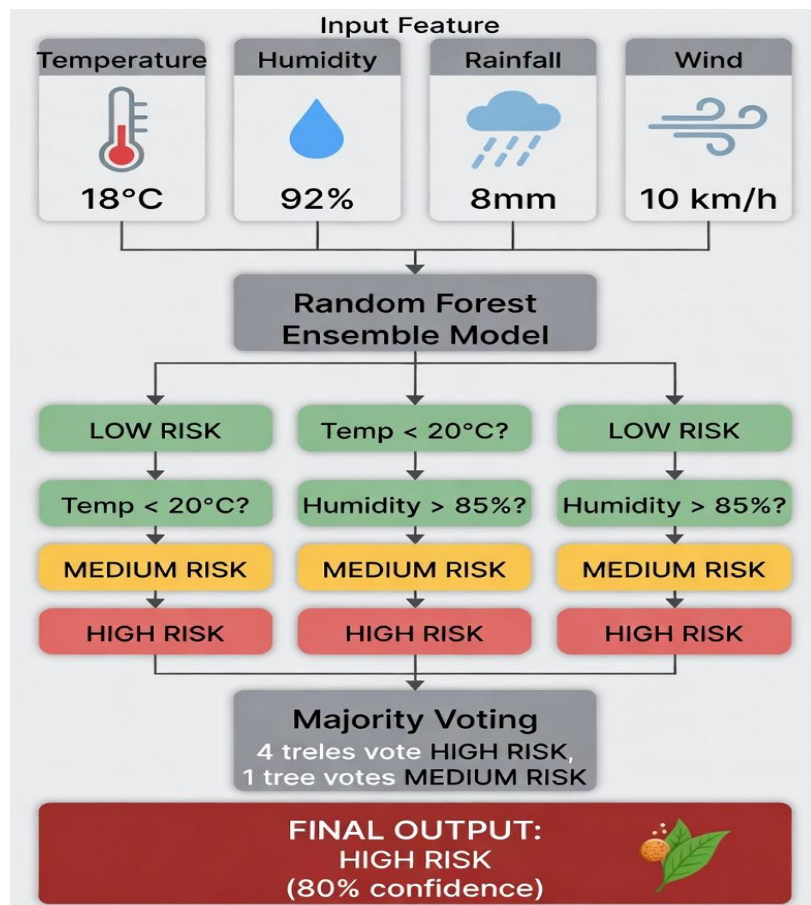


Figure 11.2.2: Random Forest Disease Prediction Model

Figure 11.2.3 above shows the circular continuous data flow of the weather intelligence system. At the top, Weather APIs box shows cloud icons for Google, OpenWeather, and Visual Crossing with arrow pointing right labeled "Real-time data every 5 min" to Data Processing box with gear icon labeled

"Validation and Cleaning". Arrow points down to ML Models box with AI brain icon labeled "LSTM + Random Forest Prediction" showing "Daily Briefing 6 AM, SMS Alerts". Arrow points right to Estate Manager box with person icon at laptop labeled "Cultivation Tasks". Arrow points down to Field Actions box with tea plantation icon, water droplet for irrigation, and spray bottle showing cultivation tasks. Arrow points left to Outcome Recording box with clipboard and checkmark icon labeled "Actual Results (yield, quality, disease)". Arrow points up to Model Retraining box with circular refresh icon labeled "Monthly Learning Update". Arrow points right back to Weather APIs completing the circle, creating a continuous learning feedback loop

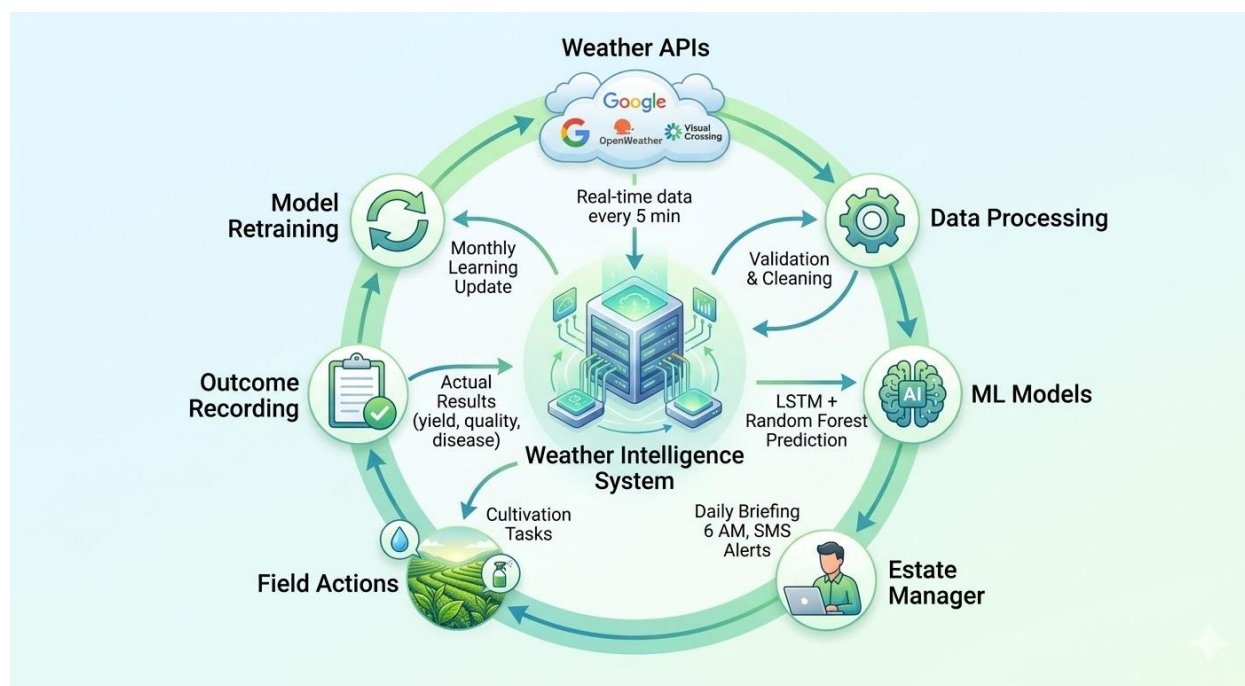


Figure 11.2.3: System Data Flow Diagram

Figure 11.2.4 above shows the five-step automated process for generating and delivering daily briefings. Step 1 labeled "COLLECT LATEST FORECAST DATA" shows clock displaying 5:30 AM and three Weather API boxes with sun, cloud, and rain icons connected by arrows to a central processing node. Step 2 labeled "ML MODEL PREDICTIONS" shows PostgreSQL database cylinder icon connecting to LSTM Model box with brain icon. The model outputs three small graphs labeled Trend, Mini Data, and Data graph showing prediction analytics. Step 3 labeled "GENERATE BRIEFING REPORT" shows a document icon with tea leaf logo containing three visible sections: Priority Actions numbered 1, 2, 3 with checkboxes, Critical Alerts section with warning triangle icon, and Quality Forecast section showing star ratings and upward trend arrow. Step 4 labeled "DELIVERY CHANNELS" shows the briefing splitting into three paths: Email Delivery with envelope icon, SMS Alert with mobile phone icon displaying "SMS", and Push Notification with smartphone icon displaying "PUSH". Step 5 labeled "MANAGER REVIEWS AT 6:00 AM" shows happy estate manager icon holding mobile device displaying briefing with time showing 6:00 AM. Green arrows connect all steps from left to right with numbered circles 1 through 5 at the bottom showing the workflow sequence.

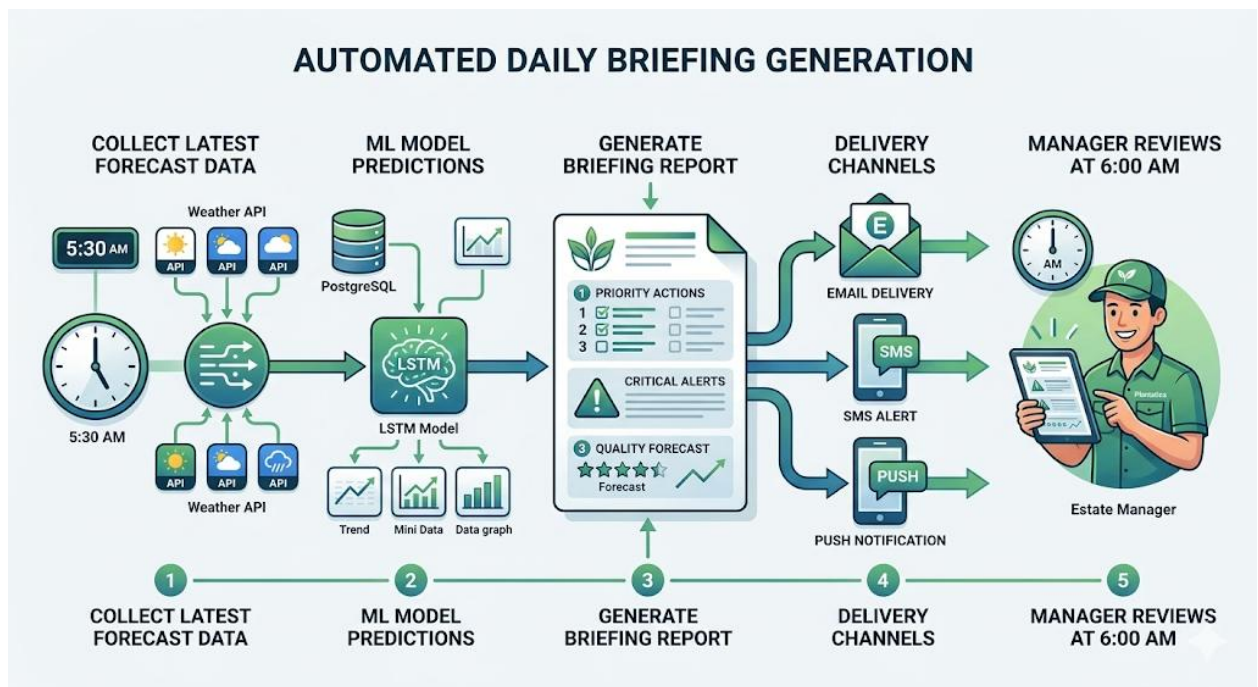


Figure 11.2.4: Automated Daily Briefing Generation Workflow

11.3 Preliminary UI Mockups

Figure 11.3.1 above shows the main web dashboard interface. The top header displays "Tea Weather AI" logo with tea leaf icon on dark green background at hex color 2E7D32. Navigation menu shows Dashboard, Forecasts, Plantation Info, Data Export, and Account Settings with user icon. Below the header, page title shows "WEATHER AI | Ceylon Estate Dashboard". The main content area contains six card-style widgets arranged in a grid. Top left card titled "CURRENT: 23°C" displays large sun icon, green checkmark, and text "Good Growing Index" with details "Humidity: 72%, Rain (24hr): 1.5mm, Wind: 10km/h SE". Top right card titled "OPERATIONS CHECKLIST" shows three tasks: "1. Harvest Sector C - Completed" with green checkmark, "2. Fungicide Spray Sector A - Needs Action" with orange warning triangle, "3. Soil Moisture Sensor Check - Check Urgently" with yellow warning. Top far right card titled "QUALITY AND YIELD FORECAST - Next 7 days" displays bar chart with green bars showing star ratings: MON 5.0 stars, TUE 2.0 stars, WED 4.0 stars, THU 4.5 stars, FRI 3.0 stars, SAT 1 star, SUN 3.0 stars. Bottom left card titled "DISEASE RISK LEVEL" shows two semicircular gauge meters. Left gauge "Blister Blight Risk" displays needle at 85% in red zone labeled "HIGH RISK". Right gauge "Red Spider Mite Risk" displays needle at 20% in green zone labeled "LOW RISK". Bottom middle card titled "WEATHER TRENDS | 7 DAY" with "Hover for detail" note shows combination chart with green area fill representing "Temperature (°C)" line ranging from 12°C at Mon to 22°C at Fri, and blue vertical bars representing "Rainfall (mm)" with values around 4mm Mon, 6mm Tue, 11mm Wed, 6mm Thu, 20mm Fri, 12mm Sun. Bottom right card titled "DAILY LOG" with "Hover for detail" shows "Day's Total logs - 110.11" with two columns "Data points: 13" and "Data points: 30", followed by "Estate Performance Summary: Estate Savrage Completed" and "Estate Parrillance Summary: Estate Performance".

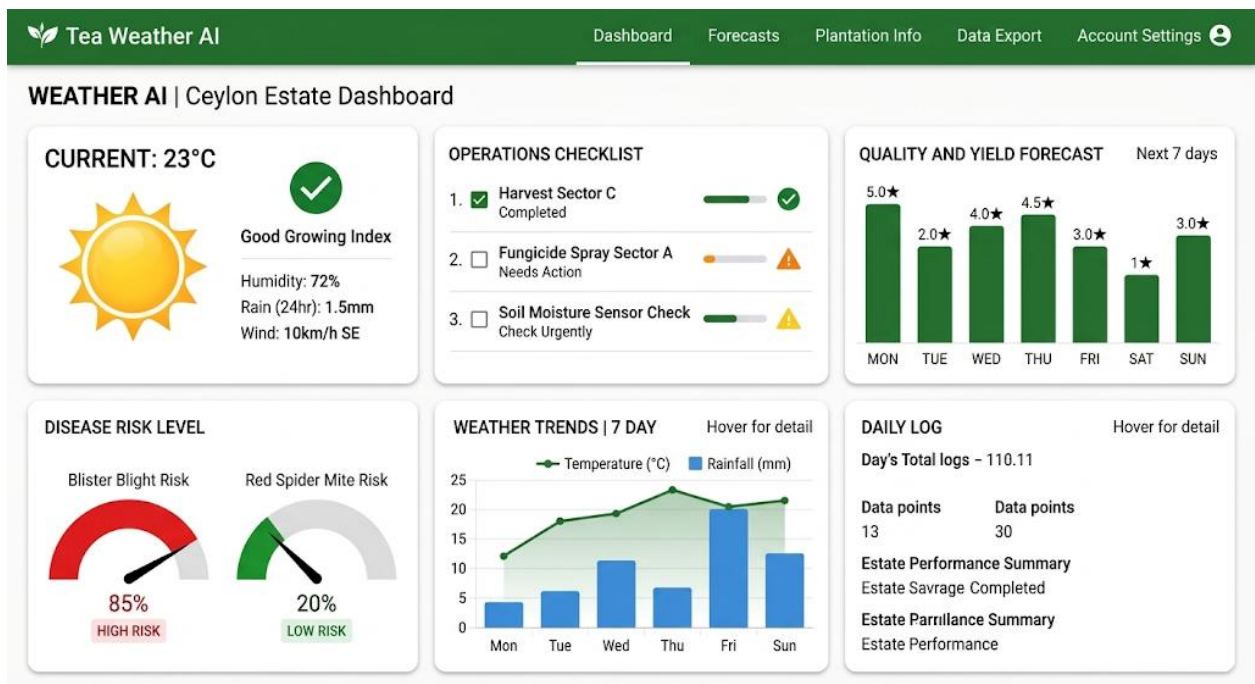


Figure 11.3.1: Web Dashboard Home Page

Figure 11.3.2 above shows the mobile application daily briefing screen on iPhone-style smartphone mockup with rounded corners and notch at top. The screen displays green header bar at hex color 2E7D32 showing back arrow on left, title "Daily Briefing - Jan 12, 2026" in center, and weather icon showing partly cloudy sun on right. Below header, location banner shows green map pin icon, text "Nuwara Eliya Estate - Excellent Conditions" and four gold stars. Main content area shows white background with multiple collapsible sections. First section "Current Conditions" with downward chevron shows sun and cloud icon "22°C → 26°C (optimal ✓)", water droplet icon "Humidity: 75% (safe range ✓)", and cloud icon "Rain: 0mm forecast". Second section "Priority Actions" displays three color-coded priority cards. Priority 1 card has red background with white text "HARVEST TODAY", followed by green checkmarks and details "Sections: A, B, C (25 hectares)", "Expected yield: 250 kg premium", "Quality: EXCELLENT", with green button "START TASK". Priority 2 card has yellow background "IRRIGATION REQUIRED" with green checkmarks, details "Sections: D, E (Evening, 4-6 PM)", "Amount: 15mm water", with blue button "SCHEDULE". Priority 3 card has green background "FERTILIZATION - WAIT" with green checkmark "Postpone to Jan 15 (rain forecast)". Third section shows red banner "Critical Alerts" containing red alert card "BLISTER BLIGHT RISK - Jan 14-16" with text "Apply fungicide TODAY" and red button "VIEW DETAILS". Bottom of screen displays large green floating action button with checkmark icon

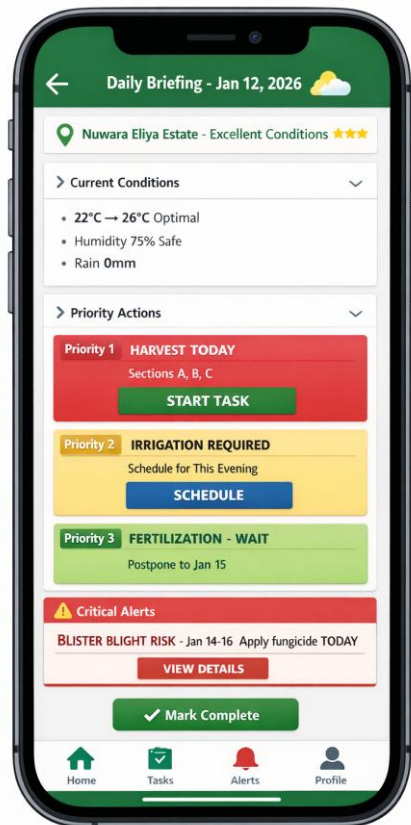


Figure 11.3.2: Mobile Daily Briefing Interface

"Mark Complete" and navigation tabs showing Home icon with "Home", checklist icon "Tasks", warning bell icon "Alerts", user profile icon "More", and three dots "More".

11.4 Risk Analysis

Table 11.4.1 below shows the risk analysis for the Tea-Specific Weather and Climate Intelligence System project, including risk descriptions, probability, impact, and mitigation strategies.

Table 11.4.1: Project Risk Analysis

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R1	Weather API service downtime or data unavailable	Medium	High	Use multiple backup APIs (3 sources: Google, OpenWeather, Visual Crossing); implement automatic switch to backup; store recent data for 24-hour operation without new API calls
R2	LSTM model learns training data too well, poor performance on new weather patterns	Medium	High	Use techniques to prevent overlearning (dropout layers, early stopping); check on separate 15% validation set; watch prediction accuracy monthly and retrain if performance drops greater than 10%
R3	Not enough or unbalanced training data for disease prediction (more "no disease" days than "disease outbreak" days)	Medium	Medium	Use SMOTE to balance classes; collect more outbreak data from Tea Research Institute historical records; use weighted loss function
R4	Limited access to tea estates for pilot use and outcome data collection	High	High	Build partnerships with 3 estates early in March 2026; offer free 6-month trial in exchange for data sharing; work with Tea Research Institute for introductions
R5	Users (estate managers) do not trust AI recommendations and ignore system guidance	Medium	High	Provide clear explanations for predictions; show confidence scores; allow manual change with feedback; show accuracy with pilot results; involve users in checking process

R6	Weather forecasts wrong beyond 3-4 days, affecting long-term planning	High	Medium	Clearly tell users prediction confidence decreases with time; focus daily briefings on 1-3 day actionable tasks; use probability forecasts like "60% chance of rain" rather than yes or no predictions
R7	System performance issues during high-traffic periods (multiple users at same time)	Low	Medium	Use cloud auto-scaling (AWS EC2); use Redis caching for frequent queries; optimize database queries; load testing before use
R8	Data privacy concerns from tea estates sharing production data	Low	High	Remove all estate-identifying information in research outputs; sign agreements with partner estates; use role-based access control; follow Sri Lankan data protection laws
R9	Budget goes over due to higher-than-expected API costs or cloud hosting fees	Medium	Medium	Watch API usage weekly; set up cost alerts at 75% budget; use API caching to reduce calls; discuss annual contracts for discounts; maintain 15% extra budget
R10	Poor internet in remote tea estates preventing real-time access	Medium	Medium	Design system to work offline for 24-48 hours using stored data; use progressive web app with offline abilities; provide SMS backup for critical alerts
R11	Resistance to technology from older estate managers used to traditional methods	Medium	Medium	Provide on-site training and support; show quick wins (time savings, cost reductions); start with early supporters as champions; design simple, easy interfaces
R12	Changes in weather patterns due to climate change making historical data less useful for predictions	Low	High	Use continuous learning system that gives more weight to recent data; watch model drift and retrain every three months; include climate trend adjustment factors; record performance changes over time

According to Table 11.4.1, the highest priority risks are R4 related to estate access, R5 related to user trust, and R6 related to forecast accuracy beyond 3-4 days. These require early attention and strong mitigation strategies. Medium priority risks include R1, R2, R3, R9, R10, and R11 which need regular

monitoring. Lower priority risks R7, R8, and R12 still require consideration but are less likely or have longer time horizons.

11.5 Ethical Documents

Ethical Research Compliance Statement

The proposed research titled "Tea-Specific Weather and Climate Intelligence System" will follow ethical research standards throughout the data collection, analysis, system development, and use phases.

Informed consent means any weather-related information or tea cultivation outcome data collected from tea estates will be obtained with clear written permission from estate management. Estate managers, supervisors, and workers will be informed about the purpose of the research which is developing AI-based weather intelligence for tea cultivation, how collected data will be used for training machine learning models and checking predictions, their right to stop participating at any time, and how their data will be protected and made anonymous in research publications.

Before conducting surveys, interviews, or system testing at tea estates, all participants will receive an information sheet explaining the research and will provide signed agreement forms.

Data privacy and confidentiality means the research will protect the privacy and confidentiality of participating tea estates. Estate identification will replace estate names with anonymous codes like Estate A or Estate B in all research outputs, publications, and presentations. Production data including sensitive business information like exact yields, revenues, and costs will not be shared publicly. Aggregated reporting means only combined statistics across multiple estates will be published such as "average yield improvement of 15%" and never individual estate performance. Personal information means no personally identifiable information about estate workers or managers will be collected or stored.

Responsible use of data means weather data from publicly available APIs including Google Weather and OpenWeather will be used following their terms of service. Outcome data including tea cultivation outcomes like yield, quality, and disease records will be used only for academic research purposes and model training. Attribution means proper citations and acknowledgments will be provided for all external data sources, research papers, and institutional datasets used.

Data security means all collected data will be stored in encrypted databases using AES-256 encryption standard. Access control means only authorized research team members will have access to raw estate data with role-based permissions limiting access by data sensitivity. Secure transmission means all data transfers between system parts will use TLS encryption protocols. Backups include encrypted daily backups with 30-day keeping period with backups stored in secure cloud storage with access recording.

Ethical AI and algorithm fairness means no bias where the system will provide equal service and recommendation quality to all tea estates regardless of size, location, or ownership. Transparency means prediction methods and data sources will be clearly documented and explainable to users. Human oversight means system provides recommendations not mandatory commands with estate managers

retaining final decision-making authority and ability to change any system suggestion. Accountability means all predictions and recommendations will be recorded with timestamps for checking and accountability.

Environmental and social responsibility means the system promotes environmental sustainability through water saving where precision irrigation reduces usage by 25-35%, pesticide reduction where targeted disease prediction reduces unnecessary spraying by 15%, and climate adaptation helping estates adjust to changing weather patterns. Worker safety includes early weather hazard warnings for thunderstorms, frost, and heat waves protecting field workers from dangerous conditions. Livelihood support means improved plantation productivity supports rural employment and tea farming communities.

Ethical research conduct means no fabrication where all data collection, model training results, and performance measures will be accurately reported without changing. No plagiarism means all literature review, methodology references, and borrowed concepts will be properly cited using IEEE format. Conflicts of interest means the research is purely academic with no financial relationships with weather API companies, tea estate corporations, or competitors that could bias results.

Participant rights include voluntary participation where tea estates participate voluntarily in pilot testing and data sharing. Right to withdraw means estates can withdraw from the study at any time without penalty. Access to findings means participating estates will receive summary reports of research findings. Feedback channels mean estates can provide feedback on system performance and recommendations.

Ethical review and approval means this research will follow the ethical research guidelines of the Sri Lanka Institute of Information Technology and will make sure that the system supports sustainable agricultural practices without causing harm to individuals, organizations, or the environment. If required by SLIIT research ethics committee, formal ethical review approval will be obtained before starting field data collection.

Commitment to responsible innovation means the research team commits to developing technology that empowers tea estate managers with better decision-making tools, reduces environmental impact through resource optimization, supports climate resilience in Sri Lankan tea cultivation, maintains transparency and accountability in AI-driven recommendations, and protects the privacy and interests of all stakeholders.